

Non-exonic read content reorganises lncRNA networks and masks tumour cell programmes in renal cancer

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Abstract

Co-expression networks are used to find prognostic lncRNA programmes, but lncRNA quantification depends on a library's unprocessed RNA content. In renal cancer (1,062 patients), the leading lncRNA axis tracked the non-exonic (non-feature) read fraction, a routine alignment metric, at Spearman rho 0.90. Without adjustment, most of the lncRNA network merged into one module. Adjusting for the metric preserved the agreement between bulk and single-cell tumour-normal differences. Across 25 TCGA cancer types, under rules fixed in advance, 9 had a lncRNA-specific axis and 4 network collapse, and intronic lncRNAs tracked the metric more

than intergenic ones in all 25. The metric had no consistent association with survival (pooled hazard ratio 1.00). Five renal modules survived adjustment. All kept their direction of effect in an external cohort and three localised to malignant or epithelial cells. Together they added no discrimination over TNM and grade.

Introduction

Prognostication in renal cell carcinoma rests on pathological stage, nuclear grade and composite scores built from them. The Mayo Clinic SSIGN score combines stage, size, grade and coagulative necrosis to predict cancer-specific survival after radical nephrectomy¹. The Leibovich score predicts metastatic progression from similar variables², and the UCLA Integrated Staging System combines stage, grade and performance status³. These routinely recorded variables are the comparator a molecular marker has to beat.

Many transcriptomic signatures, increasingly built from long non-coding RNAs (lncRNAs), have been reported to improve on these scores. Six lncRNA signatures illustrate the pattern, all derived from the same TCGA-KIRC expression matrix under different biological labels: nine lncRNAs with a survival nomogram⁴, eight autophagy-related⁵, five m6A-related⁶ and four cuproptosis-related lncRNAs⁷, a pyroptosis-related model⁸ and six disulfidptosis-related lncRNAs⁹. They share the discovery cohort and modelling strategy, and none states RNA integrity, mapping rate or per-sample sequencing quality control in its Methods.

Such signatures have been questioned on statistical grounds. When 28 published clear-cell renal cell carcinoma biomarkers were evaluated systematically in 350 TCGA patients, only tumour stage and the ccB expression signature remained independent predictors after multivariable adjustment¹⁰. In breast cancer, most randomly drawn gene signatures are associated with outcome once they correlate with a dominant biological axis such as proliferation¹¹.

A separate problem is technical. The fraction of reads outside annotated exons varies systematically between RNA-seq libraries, largely with extraction method and library selection¹², which is why the exonic rate is a routine per-sample quality metric¹³. RNA degradation widely distorts expression estimates, and covariate adjustment removes much of this distortion where normalisation does not^{14,15}. In TCGA RNA-seq, library size, tumour purity and processing plate affect gene co-expression and expression-survival associations¹⁶. Technical confounders inflate co-expression network edges, and removing leading principal components before network inference reduces false connections¹⁷. Such correction can also remove biology, a risk not yet tested against single-cell or cell-type ground truth. How far a single routine alignment metric carries the co-expression structure and apparent prognostic signal of a whole gene class has not been quantified.

Non-exonic reads should matter most for lncRNAs, which are low in abundance, often retained in the nucleus and often inside or opposite protein-coding loci, where unspliced and intronic reads accumulate¹⁸. Among the lncRNAs analysed, 58.3% overlap or sit opposite a protein-coding gene (Supplementary Table 1d), and intron retention is widespread in cancer transcriptomes¹⁹. Apparent lncRNA expression should therefore depend on a library's unprocessed RNA content, and a co-expression network built on it should inherit that dependence before any survival model is fitted.

To measure this dependence and its consequences, we assembled four renal cancer cohorts, processed through one Genomic Data Commons STAR-Counts pipeline, and four single-cell cohorts without outcome data. One renal cohort used a different library protocol. We related the leading lncRNA axis to the alignment metrics and network structure of the same libraries, then residualised expression on those metrics before defining modules. We tested modules and

published signatures for prognostic value with and without adjustment, and asked whether surviving modules replicate, transfer across protocols, localise to cells and add to TNM with grade. Finally, we tested the correction against single-cell ground truth and measured the axis and network reorganisation across 25 TCGA cancer types under decision rules fixed in advance.

Results

Four cohorts differ in library protocol and in non-exonic read content

To relate lncRNA structure to library content, we first defined four renal cohorts and compared their library protocols and alignment metrics. From 614 TCGA-KIRC STAR-Counts files we excluded 73 non-primary-tumour samples, 5 failed libraries, 4 duplicate aliquots and 4 with non-positive or missing survival time, leaving 528 patients with 173 deaths and a reverse Kaplan-Meier median follow-up of 1,657 days (Fig. 1a, Table 1). Nodal status was explicitly assessed in 259 of 528 patients (17 node-positive) and M category in 494 (79 M1). Two stage IV patients without T4 disease or an explicit M category were reconciled to M1, giving 81 M1 patients of 528. Models requiring the full clinical covariate set used 520 patients with 172 deaths.

The CPTAC-3 validation cohort kept 200 patients with 32 deaths from 353 primary-tumour files after removal of duplicate aliquots, non-clear-cell cases and missing survival times. The subtype cohorts were 269 TCGA-KIRP patients and 65 TCGA-KICH patients with 9 deaths, and the 263 TCGA-KIRP patients with a recorded T category had 40 deaths (Fig. 1b, Table 1, Supplementary Data 1).

The TCGA libraries are poly(A)-selected and the CPTAC-3 libraries ribo-depleted total RNA.

The non-feature read fraction is the percentage of reads that STAR assigns to no annotated feature in its gene-counts summary (Methods). It measures a library's non-exonic, largely pre-

mRNA read content and, with the multimapping fraction and depth, is one of three alignment metrics used throughout. Its median was 4.6% in TCGA-KIRC (interquartile range 3.1 to 6.8), 40.5% in CPTAC-3 (38.4 to 42.9), 3.9% in TCGA-KIRP (2.7 to 5.3) and 3.1% in TCGA-KICH (2.6 to 3.5) (Fig. 1c), an 8.7-fold difference between the validation and discovery cohorts. Networks were built in the discovery cohort from the 12,000 protein-coding and 3,442 lncRNA genes that passed filtering, each matrix first residualised on the non-feature fraction, the multimapping fraction and log10 assigned reads. The protein-coding network (506 samples, soft power 18) resolved 13 modules, with 4,069 genes (33.9%) unassigned. The lncRNA network (511 samples, soft power 10, fixed by an outcome-blind sweep) resolved 16 modules, with 966 genes (28.1%) unassigned (Supplementary Fig. 1). Each of the 29 module eigengenes is a rotation fitted once on the residualised discovery matrix and reproduces the WGCNA eigengene at $r = 1.000$ (Supplementary Data 2).

The leading lncRNA expression axis tracks non-exonic read content

We next asked whether the main source of lncRNA variation follows the non-feature fraction, as dependence on unprocessed RNA content would predict. The leading principal component of the observed lncRNA matrix explained 20.0% of total variance, 2.5 times the second (8.1%). It correlated with the non-feature read fraction at Spearman ρ 0.90 ($p = 1.4 \times 10^{-186}$, $n = 511$, Fig. 2a) and only weakly with depth (-0.13) and the ESTIMATE stromal and immune scores (-0.16 and 0.04). The leading protein-coding component (13.7% of variance) does not track the metric (ρ 0.05, p 0.30, $n = 506$, Fig. 2b). It tracks the multimapping fraction instead (-0.51), and its fourth and fifth components track the metric at 0.61 and 0.66.

At gene level the effect is concentrated in lncRNAs. The median absolute correlation with the metric was 0.37 across the 3,442 lncRNAs, with 56.8% of genes above 0.3, against 0.13 and

18.8% across the 12,000 protein-coding genes (Fig. 2c). The exception is the protein-coding red module (473 genes). Its observed eigengene correlates with the metric at ρ 0.84, with a median per-gene correlation of 0.68 and 94.5% of genes above 0.3, against 0.22 in the next strongest module. It is enriched for the Gene Ontology terms RNA splicing (adjusted p 4.5e-15) and mRNA processing (8.9e-14), as a shift towards unspliced and intronic read content would predict.

Normalisation could create such an axis, and Genomic Data Commons FPKM is one route into it, because it divides by reads assigned to protein-coding genes while intronic and overlapping reads accumulate at lncRNA loci. The lncRNA read share tracks the metric at ρ 0.87 (Supplementary Table 2b). Yet the axis persists under counts per million, trimmed mean of M-values, a variance-stabilising transform and quantile normalisation of the same 511 libraries, with the leading component tracking the metric at absolute ρ 0.93, 0.91, 0.92 and 0.88 (Supplementary Fig. 2). On the trimmed-mean scale, module-level hazard ratios shift downward for 7 of 16 lncRNA modules without adjustment and for none after residualisation (Supplementary Table 2).

The signal is carried by genes whose exons lie inside or opposite the transcribed span of a protein-coding gene. With lncRNAs classified by position against GENCODE v36, intronic and exonic-sense lncRNAs had median absolute correlations with the metric of 0.53 and 0.52. The intergenic and divergent classes, which overlap no protein-coding gene, both had 0.25. The class difference (Kruskal-Wallis chi-square 263.31 on 5 degrees of freedom, p 7.7e-55) survives adjustment for mean expression and log10 gene length (F 54.80, p 6.2e-55), although other gene properties could still contribute (Supplementary Table 1).

To test whether the axis requires tumour tissue, we projected the fixed discovery direction onto the 72 matched TCGA-KIRC normal-kidney libraries (72 patients, 71 in the discovery set). It correlates with the metric at $\rho = 0.86$ ($p = 8.9 \times 10^{-22}$), and the projected protein-coding direction does not (0.09, Supplementary Fig. 3). The unpaired tumour-normal difference in the metric (1.73 percentage points) largely reflects plate confounding, falling to 0.15 points on shared plates and within patient (Supplementary Table 3).

Across cohorts, the projected discovery direction correlates with the non-feature fraction at $\rho = 0.90$ in TCGA-KIRC tumours, 0.89 in TCGA-KIRP, 0.86 in TCGA-KIRC normal kidney and 0.59 in TCGA-KICH, and at 0.55 and 0.43 in CPTAC-3 tumours and normal kidney (Fig. 2d, Supplementary Table 4). The correlation is graded by library protocol, and the axis transfers to ribo-depleted libraries at reduced strength. Its lncRNA specificity is a property of poly(A)-selected libraries. Against the projected protein-coding direction the contrast is 0.90 to 0.06 in TCGA-KIRC tumours but 0.55 to 0.29 in CPTAC-3, because retaining unprocessed RNA raises intronic content for every gene.

The axis reorganises the lncRNA co-expression network

An axis this strong could also determine network structure, so we rebuilt the lncRNA network on the 511 discovery libraries with identical parameters but without adjustment. It resolves into 3 modules instead of 16. Its largest module holds 1,982 of 3,442 genes (57.6%), against 438 (12.7%) in the adjusted network, and its eigengene correlates at $r = 0.98$ with the observed leading component and 0.91 with the metric. Mean intramodular connectivity over assigned genes, each computed on its own matrix, is 70.5 against 3.3 after adjustment. The two partitions share almost no structure (adjusted Rand index 0.062 over all genes, 0.047 over the 1,722 genes assigned in both, Supplementary Table 5).

The largest unadjusted module, hereafter the dominant module, absorbs the positional classes that carry the metric's signal. It holds 71.5% of intronic and 69.8% of exonic-sense lncRNAs, against 43.7% of intergenic lncRNAs and 57.6% overall (false discovery rate $8.7e-17$ for intronic, $1.8e-29$ for intergenic). The adjusted lncRNA black, green and turquoise modules lie almost wholly inside it (99.5% to 100% of their genes), whereas the unadjusted network left 86.1% of greenyellow and 63.4% of blue unassigned. Two of the three prognostic lncRNA modules were therefore largely invisible before adjustment, and the third, turquoise, had been part of the dominant module (Supplementary Table 5).

Across TCGA the mechanism is general and varies in strength

To test whether the renal finding is specific to one dataset, we repeated the measurements in every TCGA project under decision rules fixed, committed and hashed before any non-renal project was analysed (Methods). Of 11,505 STAR-Counts files from 33 projects, 10,722 libraries were retained, and 25 projects had at least 100 retained primary tumours (Supplementary Table 6). Through the pan-cancer pipeline, TCGA-KIRC reproduced the discovery axis (ρ 0.91) and the collapsed network (3 modules, 57.5% of genes in the largest, adjusted Rand index 0.06).

The leading lncRNA component tracked the non-feature fraction at absolute ρ of at least 0.5 in 19 of the 25 cancer types (median 0.72, against 0.28 for the protein-coding component, Fig. 3a).

The locked rule for a lncRNA-specific axis, which adds a margin of 0.3 over the protein-coding component, was met in 9: KIRC, LUAD, LUSC, OV, STAD, READ, PAAD, COAD and LGG.

Of the other ten, five (GBM, SKCM, KIRP, SARC and BLCA) had a protein-coding component that also reached absolute ρ of at least 0.5, and the rest missed the margin, THCA and ESCA

narrowly. The following comparisons are descriptive and outside the locked rules. At gene level, lncRNAs tracked the metric more than protein-coding genes in 22 of 25 types, and intronic more

than intergenic lncRNAs in all 25 (median 0.35 against 0.19, six-class Kruskal-Wallis p at most
 1.3e-15 in every type, Fig. 3b). In 13 normal-tissue sets with at least 20 libraries, the lncRNA
 component reached absolute ρ of at least 0.5 in 8 and met the full rule in 2. In TCGA-KIRC
 normal kidney the projected discovery direction tracks the metric at 0.86 but the samples' own
 leading component only at 0.33, so the axis is present without dominating the variance.
 Network collapse is the extreme form of the effect. The locked rule for a reorganised network
 requires a largest observed module holding at least 40% of genes, with an eigengene tracking the
 metric at absolute r of at least 0.7. It also requires an adjusted Rand index below 0.2 between
 observed and adjusted partitions. The rule was met in 4 of 25 types (GBM, KIRC, SKCM and
 KIRP), with LUAD just outside (44%, 0.21) (Fig. 3c). Short of collapse, the largest observed
 module tracked the metric at absolute r of at least 0.7 in 17 of 25 types, and the partitions agreed
 only moderately (median adjusted Rand index 0.31).
 To see why the effect varies in strength, we asked whether it follows the within-cohort variability
 of the metric, in an exploratory analysis specified after the locked rules were applied. Spread was
 measured as the median absolute deviation of the log non-feature fraction. Across the 25 cancer
 types, spread correlated with lncRNA axis strength (Spearman ρ 0.73, false discovery rate
 6.1e-04) and with the share of genes in the largest observed module (0.69, 0.0016) (Fig. 3d,
 Supplementary Table 6d). The first association held for all four spread measures (0.63 to 0.73),
 and the second weakened with the standard deviation (0.38, 0.081). No spread measure predicted
 the adjusted Rand index, and GBM, with by far the most variable libraries, contributes much of
 the pattern.

**The metric is structured by batch, prognostic in renal cancer and not consistently so across
 cancer types**

206 Studies that treat lncRNA scores as biology would read the axis as a prognostic marker, so we
 207 tested it, and the metric itself, in survival models. Hazard ratios are per standard deviation of the
 208 exposure or eigengene throughout. In nested Cox models (503 patients, 163 deaths), the axis
 209 gave 1.31 alone (95% confidence interval 1.12 to 1.53) and 1.34 to 1.39 after adding clinical
 210 terms, ESTIMATE scores, RNA integrity and slide necrosis, so it appears prognostic
 211 independently of the recorded clinical variables. The three STAR metrics reduced it to 1.08 (0.81
 212 to 1.43, p 0.61) with a variance inflation factor of 4.5, so collinearity contributes to the collapse.
 213 Stratifying the fullest model by sequencing plate gave 0.90 (0.62 to 1.29) (Fig. 2e,
 214 Supplementary Table 7a).
 215 Tested directly, the non-feature fraction is itself prognostic. In 520 patients (172 deaths) it gave
 216 1.28 alone (1.13 to 1.46, p 1.2e-04) and 1.36 after clinical adjustment (1.20 to 1.55, p 3.0e-06),
 217 or 1.44 on the log10 scale (1.23 to 1.68) (Supplementary Table 8c). Adding ESTIMATE scores
 218 and stratifying by tissue source site gave 1.45 (1.25 to 1.69, p 1.6e-06). In the 503 patients with
 219 RNA integrity recorded, it remained prognostic within sequencing plate at 1.23 (1.02 to 1.50, p
 220 0.035) after adjustment for RNA integrity, slide necrosis, the multimapping fraction and depth,
 221 which removes the axis (Supplementary Table 7b). In CPTAC-3 it was not prognostic, with a
 222 reversed point estimate of 0.76 (0.45 to 1.27, p 0.29, 199 patients, 32 events). Two explanations
 223 fit, and these data cannot separate them: 32 events give little power, and under that cohort's
 224 protocol the metric has a different distribution and possibly a different meaning.
 225 To see what the metric reflects, we partitioned its variance. It is structured by batch: sequencing
 226 plate explains 26.6% of its variance (p 6.5e-27), Biospecimen Core Resource batch 24.6% and
 227 tissue source site 11.1%, and the same three explain 26.2%, 26.0% and 11.6% of variance in the
 228 axis (Fig. 2f, Supplementary Table 7d). Plate medians range from 2.6% to 7.1% across the 14

plates with at least ten patients. RNA integrity, recorded for 527 of the 528 patients, correlates with the metric at -0.25 ($p = 9.4 \times 10^{-9}$) and explains 6.9% of its variance alone and 36.2% jointly with plate and site (Supplementary Table 7c, e). Slide percent necrosis is uninformative (median zero, correlation -0.05, $p = 0.28$), and no clinical variable is associated with the metric (smallest $p = 0.21$, Supplementary Fig. 4, Supplementary Table 9f).

Biological correlates point the same way. In 501 patients, a block of annotated tumour biology (hypoxia metagene, proliferation score, ESTIMATE scores and slide tumour-nuclei percentage) explains 6.1% of the metric's variance, against 36.8% for RNA integrity, plate and site. Blocks built from observed expression explain more but cannot be identified as biology, because random gene sets of the same sizes reach 61.0%. Singly, proliferation is uncorrelated with the metric ($\rho = -0.01$) and hypoxia correlates in the unexpected direction (-0.14). Stratifying the clinically adjusted model by plate moves the hazard ratio from 1.37 to 1.24 (1.04 to 1.48) for the metric and from 1.36 to 1.13 (0.94 to 1.36) for the axis. No biological covariate moves the metric's log hazard ratio by more than 5% (Supplementary Fig. 5, Supplementary Table 9).

Driver genotype was not associated with the metric or the axis (5% false discovery rate) in the 368 patients with open-access mutation calls, a subset with a lower metric and half the death rate of the rest. Adjusting for genotype left the metric's hazard ratio at 1.34 (1.10 to 1.63, Supplementary Table 1).

If the renal association reflected a general effect of library content on outcome, it should recur in other cancer types, so we tested the 20 eligible types with at least 20 deaths. Plate-stratified Cox models adjusted for age, sex and, where recorded, stage gave a pooled hazard ratio per standard deviation of the metric of 1.00 (0.93 to 1.08, I-squared 50%, heterogeneity $p = 0.0047$, prediction interval 0.79 to 1.26). Estimates exceeded one in 10 types, and 4 were nominally significant,

three above one (GBM, KIRP and UCEC) and one below (HNSC) (Fig. 3e, Supplementary Table 6). Without plate stratification the pooled estimate was 1.01 (0.94 to 1.08). For TCGA-KIRC, the pan-cancer pipeline (Clinical Data Resource endpoint, stage in place of T, N, M and grade) gives 1.27 (1.12 to 1.43) without and 1.13 (0.96 to 1.33) with plate stratification.

Comparison with single-cell ground truth finds no loss of tumour biology after correction

Residualisation could remove tumour biology along with artefact, so we compared its effect with single-cell data, which do not depend on bulk library preparation. In 72 TCGA-KIRC patients with tumour and matched normal libraries, the bulk tumour-minus-normal difference of 507 lncRNAs agreed with their single-cell fold change between malignant and normal epithelial cells at Spearman rho 0.775, and 92.5% of the 213 single-cell-significant lncRNAs agreed in sign (Supplementary Fig. 6, Supplementary Table 10). After correction, paired agreement fell slightly but detectably to 0.771 (-0.004, 95% interval -0.008 to -0.001). This paired contrast is insensitive, because a patient's tumour and normal libraries differ little in the correction covariates (0.13 standard deviations in the non-feature fraction). The unpaired contrast of all 532 tumours against all 72 normals, added post hoc, is more sensitive, because the gap there is 0.86 standard deviations. Agreement was 0.773 before and 0.774 after correction (change 0.001, -0.013 to 0.014), and sign agreement 93.9% and 91.5% (change -2.4 points, -5.6 to 0.9). The correction therefore neither improved nor detectably degraded agreement with single-cell biology.

Within tumours, residualisation removes by construction the correlation of any lncRNA score with the metric. The informative quantity is therefore how far a score's coherence with a protein-coding malignant-marker score exceeds that of lncRNA sets that single-cell data classify as non-specific. A score from the 38 single-cell tumour-specific lncRNAs, fixed before any bulk step,

correlated with the marker score at 0.55 (0.48 to 0.62) before and 0.48 (0.41 to 0.55) after correction. Abundance-matched non-specific sets gave 0.37 and 0.28, an excess of 0.19 before and 0.20 after. All sets lose some coherence because the marker score itself correlates with the metric at 0.22. Three expectations recorded in the analysis script before it was run were not met. Paired agreement fell slightly, and the non-specific sets were not neutral. The 15 candidates set aside in bulk as technically sensitive lost their correlation with the metric (0.50 to 0.84 before, - 0.08 to 0.11 after) but had no room to show recovery, because four of the five with a normal-epithelium estimate already agreed with single-cell direction before correction.

The conclusions do not depend on the correction method

To test whether these conclusions depend on the correction method, we repeated the analyses on the same 528 tumours with RUV-III using pseudo-replicates of pseudo-samples¹⁶. The conclusions were unchanged (Supplementary Table 11). Pseudo-replicates were built within ccA/ccB and stage groups, across plates and across extremes of the metric and depth. RUV-III removed the link between the leading lncRNA component and the metric (rho 0.05, against 0.06 after residualisation and 0.91 observed) and prevented network collapse (largest module 14% of genes, against 14% and 58%). All five prognostic modules stayed associated with survival at similar hazard ratios. The two corrections agreed only moderately on module membership (adjusted Rand index 0.23). RUV-III also removed more single-cell-confirmed tumour signal. The tumour-specific lncRNA score correlated with the malignant-marker score at 0.28 after RUV-III and 0.48 after residualisation, and its excess over abundance-matched non-specific sets drawn for this comparison was 0.11 and 0.22. RUV-III's controls may capture some tumour-content variation that residualisation keeps by design.

Adjusting for the alignment metrics changes which modules appear prognostic

A technical axis matters only if it changes what a study concludes, so we next asked whether adjustment, with module membership held fixed, changes which modules appear prognostic. Across the 29 modules, with the clinical model and Benjamini-Hochberg control within each network, 15 were significant with eigengenes from observed expression (11 of them lncRNA modules), ten with expression residualised on the three STAR metrics (Fig. 4a) and 3 with observed eigengenes plus the three metrics as covariates (Supplementary Table 12a). Nine modules were lost on adjustment, four gained, six significant in both arms and ten in neither. The largest change is in the protein-coding red (splicing) module, from 1.45 per standard deviation (1.24 to 1.69, FDR 2.5e-05) to 1.17 (0.99 to 1.40, FDR 0.16). With RNA integrity and slide necrosis added to the clinical model, 12 of 29 are significant without adjustment and 4 with it (Supplementary Table 12c).

Applied to published work, the same test gives a more qualified answer. Six published TCGA-KIRC lncRNA prognostic signatures were re-scored on the 528 discovery tumours, with 34 of their 38 genes present and scoreable in the current annotation. Every signature correlates with the metric, five positively and the Yu signature negatively. Spearman correlations were 0.17 and 0.43 for the two with published coefficients, Hong 2022⁷ and Jiang 2019⁴, and 0.11 to 0.43 for the three whose weights were signed by the univariate Cox coefficient in the same patients, Liu 2023⁸, Gui 2021⁵ and Feng 2024⁹. Yu 2021⁶ mapped only 2 of its 5 lncRNAs and lost the gene with the largest published coefficient, so its truncated score runs opposite to the published model (-0.24) and is reported as a footnote. After residualisation every correlation lies within 0.04 of zero (all p above 0.41).

No signature is abolished by adjustment for the alignment metrics. With clinical adjustment (520 patients, 172 events), the two with published coefficients gave 1.67 (1.45 to 1.93) for Jiang and 1.75 (1.51 to 2.03) for Hong, and the three weighted in sample 1.50 to 1.63 (Supplementary Table 13). Adding the three STAR metrics changed the log hazard ratio by -2.6% and -4.9% for the first two and by -19.0% to -11.1% for the other three, and every adjusted estimate kept p below 0.001. Nor are the signatures less correlated with the metric than chance: against a sign-randomised null, three of the six sit at or above the null median (Supplementary Fig. 7). All six were derived in TCGA-KIRC, so their increments over a clinical model in this cohort (0.021 and 0.024 for the two with published coefficients) are apparent figures.

Five modules remain associated with survival after full adjustment

To find modules that stay prognostic once the alignment metrics and recorded confounders are accounted for, we used a principal specification that adds the module eigengene to age, sex, T category, nodal status, M1, ordinal grade, the ESTIMATE stromal and immune scores and the three STAR metrics. Under it, five modules remained significant at a within-network false discovery rate below 0.05, and exactly these five survive Benjamini-Hochberg control across all 29 modules of both networks (Table 2, Fig. 4b, Supplementary Data 3). Two are protein-coding (502 patients, 165 events): green (670 genes) at 0.62 (0.53 to 0.74, FDR $2.2e-07$) and purple (216 genes) at 1.28 (1.09 to 1.50, FDR 0.016). Three are lncRNA modules (504 patients, 163 events): blue (246 genes) at 0.71 (0.60 to 0.83, FDR $4.8e-04$), greenyellow (108 genes) at 1.35 (1.15 to 1.60, FDR 0.0027) and turquoise (438 genes) at 1.24 (1.06 to 1.45, FDR 0.039). The lncRNA eigengenes also track stage and grade (Fig. 4c), greenyellow at r 0.31 with ordinal AJCC stage and 0.38 with ordinal grade, and blue at -0.16 and -0.22.

The five associations do not depend on the coding of unassessed nodal and metastasis status. Alternative codings moved every hazard ratio by at most 0.026, and restriction to patients with both assessed left all five in the same direction (Supplementary Table 14). The eigengene term meets the proportional-hazards assumption in every model (p 0.45 to 0.95 for the five modules), and global-test violations lie in the clinical covariates.

Because module eigengenes differ by sex (8 of 29 at a 5% false discovery rate), we also fitted sex-specific models. Greenyellow is nominally stronger in women in each cohort where a sex-specific fit converges, at 1.97 (1.40 to 2.78) against 1.20 (0.98 to 1.47) in TCGA-KIRC, but no interaction survives multiplicity correction, so these estimates are descriptive (Supplementary Table 15).

All five modules keep direction in CPTAC-3 and the networks transfer

Associations selected in one cohort need confirmation in an independent one, so we projected the five modules onto CPTAC-3 with membership and loadings fixed in discovery and re-estimated only the module coefficient. All five kept their discovery direction (Table 2, Fig. 5a, one-sided sign test p 0.031). Evidence for individual modules is weaker. In 199 patients with 32 deaths, under the six-parameter set meeting the events-per-variable threshold of 5, lncRNA greenyellow gave 1.42 (1.07 to 1.90, p 0.016) and protein-coding purple 1.41 (1.04 to 1.92, p 0.027), neither reaching a 5% false discovery rate (0.068). Under the twelve-parameter principal set (2.7 events per variable) they gave 1.59 (1.12 to 2.25) and 1.52 (1.08 to 2.14), both at 0.042. Green, blue and turquoise kept direction without significance, so greenyellow and purple are nominally replicated.

We then projected the modules onto papillary renal cell carcinoma with a six-parameter model omitting grade, and four of the five kept direction and significance (263 patients, 40 events).

Protein-coding green gave 0.62 (0.45 to 0.85, p 0.0030), lncRNA blue 0.72 (0.52 to 0.98, p 0.040), lncRNA greenyellow 1.46 (1.10 to 1.93, p 0.0080) and protein-coding purple 2.27 (1.75 to 2.93, p 5.8e-10), and lncRNA turquoise was null at 1.06 (0.79 to 1.42, p 0.71) (Fig. 5b). Only purple differs between TCGA-KIRC and TCGA-KIRP (Cochran Q 12.38, p 4.3e-04, I-squared 92.0%), the other four giving I-squared of 0.0% to 16.0%. Chromophobe renal cell carcinoma provided 9 events for six parameters and is not interpreted (Supplementary Table 16).

To test whether the network structure transfers as well as the coefficients, we computed preservation statistics. In CPTAC-3, 13 of 16 lncRNA and 13 of 13 protein-coding modules are strongly preserved by the WGCNA permutation summary statistic²⁰ (Zsummary above 10, 100 permutations for the lncRNA and 30 for the protein-coding network). Zsummary increases with module and test-set size, so values are read only against the conventional bands. Every prognostic module is strongly preserved there: lncRNA blue 30.7, greenyellow 15.1 and turquoise 13.4, protein-coding green 61.6 and purple 52.2. In TCGA-KIRP the protein-coding modules and lncRNA turquoise are strongly preserved (34.8, 36.7 and 29.4), while lncRNA blue (8.5) and greenyellow (3.5) fall in the weak-to-moderate band. In TCGA-KICH lncRNA blue is not preserved (1.8) (Fig. 5c, Supplementary Fig. 8, Supplementary Table 17).

Surviving modules are differentiation, proliferation and malignant-cell programmes

To interpret the surviving modules, we examined their gene content, cellular origin and relations to one another. The protective protein-coding green module is a catabolic programme. Its three strongest Gene Ontology terms are small molecule catabolic process (adjusted p 4.1e-38, 85 of 613 mapped genes), organic acid catabolic process (6.2e-37) and carboxylic acid catabolic process (6.2e-37) (Fig. 6a). The adverse protein-coding purple module is a proliferation

programme, with chromosome segregation at $1.5e-44$ (Supplementary Data 4). Because these gene sets were selected by correlation, their enrichment p values serve as descriptive rankings. The green module is better described as a proximal-tubule differentiation programme than as catabolism alone. Seventeen of its top 20 hubs have a Human Protein Atlas single-cell specificity string naming proximal tubule²¹, although only LRP2 is exclusive to it. A 21-gene proximal-tubule marker score²¹, including 14 green-module genes, correlates with the green eigengene at $r = 0.84$ ($n = 506$). In a common set of 502 patients with 165 events, clinically adjusted hazard ratios are 0.69 (0.59 to 0.81) for the eigengene and 0.70 (0.59 to 0.82) for the marker score. The two cannot be separated statistically (Wald $p = 0.48$), so the module's value beyond the marker score is unresolved (Supplementary Table 18c).

The module is not stromal or immune contamination: 4 of its 670 genes overlap the ESTIMATE stromal signature against 6.4 expected, whereas protein-coding black carries 28 of 323 ($p = 1.5e-19$, Supplementary Table 18e). Nor, in discovery, is it normal-kidney admixture, as no normal library exceeds the tumour maximum on the green eigengene. In CPTAC-3, however, 6 of 168 normals do, so this argument does not extend there (Supplementary Table 18f). Consistent with ccA biology^{22,23}, the adverse hazard of ccB class after clinical adjustment (1.62, 1.16 to 2.27) disappears when the green eigengene is added (1.03, 0.67 to 1.58), a partly definitional contrast (Supplementary Table 18d).

Two of the three prognostic lncRNA modules are coupled to this programme, with signs matching the direction of risk. The protective blue eigengene correlates with green at $r = 0.91$ in TCGA-KIRC ($n = 497$), 0.92 in CPTAC-3 (199), 0.83 in TCGA-KIRP (269) and 0.71 in TCGA-KICH (65), and the adverse greenyellow at -0.57, -0.54, -0.42 and -0.18 ($p = 0.14$ in TCGA-KICH). The adverse turquoise eigengene is uncorrelated with green in every cohort (0.02, -0.04,

-0.04 and -0.06) (Fig. 6b). Blue and green are too close to separate: entered together in 493 patients with 159 events, blue carries no independent hazard (1.18, 0.79 to 1.77, p 0.41) against 0.61 for green (0.40 to 0.91, p 0.015), with variance inflation 7.5. By likelihood-ratio test, turquoise adds to green (p 6.9e-04) and greenyellow does not (p 0.19).

Gene Ontology terms of protein-coding partners, split by correlation sign, reproduce the coupling without using module membership. Blue's positive partners are enriched for small molecule catabolic process and its negative partners for extracellular matrix organization (Supplementary Fig. 9a). Greenyellow's negative partners are enriched for renal system and organic acid catabolic processes, and its positive partners for a mitotic programme (Fig. 6c). Turquoise's positive partners are enriched for RNA splicing and its negative partners for proteasomal catabolism, with no green-module overlap (Supplementary Fig. 9b).

Matched normal kidney distinguishes the three modules. Projected onto the 72 normal libraries, only greenyellow keeps its tumour-versus-normal direction after adjustment, lying 0.76 tumour standard deviations below the tumour mean (p 3.2e-11, Supplementary Table 18f). The apparent separation of blue and turquoise in normal tissue is explained by tumour-normal differences in the alignment metrics and collapses on residualisation.

Most members of the three modules are unnamed clone-based accessions (140 of blue's 246, 68 of greenyellow's 108 and 321 of turquoise's 438) (Supplementary Fig. 9c). Three greenyellow members have prior functional studies in this disease: LUCAT1²⁴ and LNCSRLR²⁵, reported as adverse like the module, and FILNC1²⁶, reported as a tumour suppressor, contrary to the module. To locate these programmes in cells, we turned to four independent single-cell cohorts^{27,28,29,30}.

Donor-level pseudobulk contrasts of malignant cells were available against normal kidney epithelium in three cohorts and against the rest of the tumour microenvironment in four

(Methods). Each module's median member log₂ fold change was compared with 2,000 random gene sets of the same gene class matched on single-cell abundance (Fig. 6d, Supplementary Table 19, Supplementary Fig. 10). Green was higher in malignant cells than in the microenvironment in all four cohorts (median log₂ fold change 2.3 to 3.6, null near zero) and lower than in normal epithelium in all three (-0.09 to -0.98), the pattern of an epithelial differentiation programme retained by tumour cells but reduced relative to normal tubule. Purple, the proliferation programme, was higher in malignant cells than in normal epithelium in all three cohorts (0.69 to 1.26) but lower than in the microenvironment in all four (-0.31 to -0.52), as expected of a programme shared with cycling immune cells. The lncRNA blue module followed green, higher than the microenvironment in all four cohorts (3.6 to 5.3, with 78% to 89% of tested members individually significant) and below the null against normal epithelium in all three, significantly in one.

The adverse greenyellow module was malignant-specific. Against normal epithelium its members shifted upwards in all three cohorts (median log₂ fold changes 1.04, 4.38 and 2.86, empirical p 0.0025, 0.0005 and 0.0005). Averaged across cohorts this is the largest shift of the 16 lncRNA modules, and none of its tested members was significantly lower in malignant cells. It was also higher than the microenvironment in all four cohorts (1.5 to 1.9). In the pooled single-cell meta-analysis, members significantly raised in malignant cells include LUCAT1 and CYTOR against normal epithelium and LNCSRLR and FILNC1 against the microenvironment.

Turquoise, the surviving module most exposed to the metric, showed only weak shifts and no malignant-specific enrichment. Its members sat slightly above the null against normal epithelium in all three cohorts (median log₂ fold changes 0.24 to 0.82, significant in one) and below it against the microenvironment in all four (significant in three). Coverage is partial because

lncRNAs are sparsely detected in single cells: 25% of greenyellow, 44% of blue and 18% of turquoise members were testable against normal epithelium in at least one cohort.

Modules add no demonstrable value over the clinical model for overall survival

A molecular marker is useful only if it improves on routinely recorded variables, so we measured incremental value over the clinical model by offering all 29 module eigengenes to a penalised Cox model, in 493 discovery patients (159 events) and 199 CPTAC-3 patients (32 events) (Fig. 7a, Table 3, Supplementary Table 20). The apparent concordance-index increment, scoring the locked model on the patients it was fitted to, was +0.021 (0.002 to 0.040, p 0.026). Under ten repeats of ten-fold cross-validation with module definitions fixed on the full cohort, a paired patient bootstrap of the repeat-averaged out-of-fold predictor gave +0.009 (-0.007 to 0.026, p 0.27). Rebuilding both networks and all loadings inside each training fold (three folds, five random fold assignments) gave +0.006 (standard deviation 0.011). Against the augmented model, which adds the ESTIMATE scores and three STAR metrics, the bootstrap increment is +0.010 (-0.007 to 0.029, p 0.22) and the fold-wise rebuild +0.012 (Table 3b).

Individual lncRNAs might carry signal that module eigengenes average out, so we also built a signature of hub lncRNAs, with the candidate pool selected inside each training fold. Over the clinical model it adds +0.007 on average across the ten repeats (bootstrap +0.020, 0.002 to 0.037, p 0.033), and over the augmented model -0.010 (bootstrap +0.003, -0.012 to 0.019, p 0.67) (Supplementary Fig. 11). The lncRNA signal such selection finds largely restates the alignment metrics already in the augmented model.

In external validation the locked model gave a concordance index of 0.759 (0.671 to 0.847) against 0.750 (0.657 to 0.843) for the clinical model, an increment of +0.009 (-0.033 to 0.048, p 0.68), or +0.004 (-0.038 to 0.043, p 0.88) with discovery-fixed standardisation (Fig. 7b,

Supplementary Table 21). Dropping each retained module in turn changed the increment by less than 0.02, and dropping all lncRNA modules left it at +0.010, so any external increment comes from the protein-coding modules.

Absolute risk and clinical utility do not favour the combined model either. In CPTAC-3 its calibration slope was 0.89 (0.58 to 1.20) against 0.97 (0.60 to 1.34) for the clinical model, and the three-year observed-to-expected ratio 0.74 against 0.75, so over-prediction in this lower-risk cohort comes from the transported baseline, not the modules (Table 3c, Fig. 7c). Brier scores favour the clinical model at one and five years and the combined model at three (Table 3d, Fig. 7d). At three years, decision-curve analysis gave a median change in net benefit of -0.004 (-0.022 to 0.006) over thresholds 0.10 to 0.40, with medians near zero in the 0.05 to 0.30 window and at five years. The curves show neither consistent benefit nor consistent harm (Fig. 7e, Supplementary Fig. 12, Supplementary Table 21).

This conclusion is specific to overall survival. For Clinical Data Resource endpoints the cross-validated increment was +0.023 for disease-specific survival (0.004 to 0.045, p 0.017, 99 patients, 99 events) and +0.014 for the progression-free interval (0.001 to 0.027, p 0.036, 146 patients, 146 events), against +0.008 for overall survival. Neither survives Benjamini-Hochberg adjustment (0.051 and 0.054 over three distinct endpoints, 0.068 and 0.072 if overall survival from the pipeline and the Clinical Data Resource counts twice). Under three-fold cross-validation the disease-specific increment falls to +0.013 whether modules are fixed or rebuilt in every fold. CPTAC-3 has neither endpoint (Supplementary Fig. 13, Supplementary Table 8).

After clinical adjustment the metric itself predicts overall survival (1.36, 1.20 to 1.55) and disease-specific survival (1.31, 1.10 to 1.55) but not the progression-free interval (1.11, 0.95 to 1.30), and the axis behaves the same way (Supplementary Table 8).

Discussion

The leading axis of lncRNA expression in TCGA-KIRC tracks a routine alignment metric, the non-feature read fraction (Spearman rho 0.90), and sets the structure of the lncRNA co-expression network. Without adjustment the lncRNA network collapses into one module holding 58% of genes. Adjusting for the three alignment metrics changes which modules exist and uncovers modules that the unadjusted network had left largely unassigned. With membership held fixed, it also cuts the significant modules from 15 of 29 to 10. The surviving modules keep the direction of their survival association in an external cohort and are preserved across library protocols, and in single cells three of them localise to malignant or epithelial cells. For overall survival they add no demonstrable discrimination over recorded TNM and grade. Across TCGA, under rules fixed before analysis, a lncRNA-specific axis occurs in 9 of 25 cancer types and the network collapses in 4, renal clear-cell among them.

Whether adjustment is appropriate depends on what the metric measures, and two readings lead to opposite conclusions. If it is pre-analytical, it indexes how the specimen was collected, stored and made into a library. If it is tissue-intrinsic, because degraded and necrotic tumour yields more unprocessed RNA and coagulative necrosis is part of the SSIGN score¹, residualising on it would remove biology. The evidence favours a large pre-analytical component. RNA integrity, plate and tissue source site explain 36.2% to 36.8% of its variance, against 6.1% for annotated tumour biology, and the axis is present in tumour-free kidney. Two further results bear on this. First, single-cell data, which do not depend on bulk library preparation, show no loss of tumour biology after correction. In the unpaired tumour-normal contrast, which the correction can alter, agreement with malignant-cell specificity was 0.773 before and 0.774 after, and a tumour-specific lncRNA score kept its excess coherence with malignant markers. This is consistent with

the correction removing pre-analytical variation and sparing biology, though it does not show that everything removed was artefact. Second, the metric has no consistent association with death across cancer types. In renal cancer it is prognostic within sequencing plate in discovery, at 1.23 (1.02 to 1.50), and weaker through the pan-cancer pipeline, which uses a different endpoint source and stage in place of TNM and grade, at 1.13 (0.96 to 1.33). Across 20 cancer types the pooled hazard ratio is 1.00 (0.93 to 1.08), with heterogeneity and nominal associations in both directions. This argues against a general marker of tumour aggressiveness while leaving room for a tissue-specific effect. Why the metric predicts death in renal cancer remains open.

The network result follows from how a signed, soft-thresholded co-expression network treats a shared axis. A variable that moves thousands of genes together raises every pairwise correlation among them, the soft threshold amplifies those correlations, and the most exposed genes, the lncRNAs lying inside or opposite a protein-coding gene, merge into one module. The positional gradient appears in every cancer type examined, so the mechanism is not specific to renal cancer. The within-cohort spread of the metric, which tracked the strength of the axis in an exploratory analysis, can be measured before any network is built. Adjusting for such covariates before building networks that include lncRNAs is established practice in other fields³¹, and the choice of adjustment is known to reshape network architecture³². Principal-component correction of TCGA correlations has so far excluded lncRNAs³³, and the TCGA lncRNA literature has not adopted covariate adjustment. Two consequences follow. First, because the dominant module holds most of the unadjusted network, the modules, hubs and guilt-by-association annotations of an unadjusted lncRNA network largely describe the metric. Second, reproducibility does not separate technical from biological structure. Turquoise lay almost wholly inside the dominant module before adjustment, and its residualised eigengene still correlates with the metric

(Spearman 0.20), yet it is strongly preserved in CPTAC-3 and TCGA-KIRP (Zsummary 13.4 and 29.4). Preservation alone does not show biology, so a check independent of library preparation is needed. The same applies to analyses that read the TCGA intronic signal as transcriptional biology³⁴.

Cell-type localisation provides that check and shows what the surviving modules represent. Protein-coding green is enriched for small-molecule and organic-acid catabolism, the metabolic reprogramming described in this disease^{35,36}. Its hubs are proximal-tubule genes²¹, and a 21-gene proximal-tubule marker score reproduces its hazard in the same patients. In single cells from every cohort, malignant cells express green far above the microenvironment and below normal epithelium, the signature of a differentiation programme retained to varying degrees by tumour cells. This explains its protective association and why it absorbs the ccA and ccB contrast²².

Purple is a proliferation programme raised in malignant cells over normal epithelium. The lncRNA blue module follows green in bulk (r 0.91) and in single cells and carries no hazard of its own once green is in the model. Greenyellow is the one lncRNA programme that is both malignant-specific and prognostic. It is the lncRNA module most enriched in malignant cells over normal epithelium, is anti-correlated with the proximal-tubule programme and is coupled to proliferation, and its members raised in malignant cells include LUCAT1²⁴ and CYTOR. In a joint model, however, it adds nothing over green (p 0.19), so its prognostic content is shared with the differentiation axis. Turquoise, the only lncRNA module that adds over green (p 6.9e-04), shows no malignant-specific enrichment and did not replicate. None of the three lncRNA modules therefore carries replicated prognostic information beyond the protein-coding programmes.

569 Applied to published TCGA-KIRC lncRNA signatures, the same test gives a mixed result. All
570 six re-scored signatures correlate with the metric, and adjusting for it abolishes none. The
571 clinically adjusted log hazard ratio moves by 2.6% and 4.9% for the two signatures that publish
572 coefficients, and by 11.1% to 19.0% for the three whose hazard direction was set in sample, a
573 figure inflated by that in-sample weighting. The data support a stronger criticism, that the
574 published increments are optimistic. The increments those signatures add to a clinical model,
575 0.021 and 0.024, are apparent figures from the cohort each was derived in. Our own modules
576 give 0.021 apparent and 0.009 cross-validated.

577 Against the strongest clinical comparator available, the modules add nothing demonstrable for
578 overall survival, with increments of 0.009 (-0.007 to 0.026) in cross-validation and 0.009 (-0.033
579 to 0.048) in CPTAC-3, and calibration, Brier scores and net benefit do not favour the combined
580 model. Hub lncRNAs selected inside each training fold add 0.020 over the clinical model but
581 0.003 over one that already contains the alignment metrics, so they largely restate those metrics.

582 The disease-specific and progression-free increments, 0.023 and 0.014, do not survive
583 Benjamini-Hochberg correction across endpoints and rest on one cohort. The comparator and
584 cohort sizes set the scope of this result. Tumour size and performance status are not recorded, so
585 the comparator is weaker than the composite scores used clinically and the reported increment is
586 an upper bound on what a fully staged comparator would leave. CPTAC-3 contributed 32 deaths
587 against the 100 to 200 recommended for external validation of a survival model³⁷, and in
588 discovery the smallest increment cross-validation could separate from zero was about 0.024. The
589 result therefore bounds the gain in these cohorts without showing it to be zero. Overall survival
590 counts deaths from any cause, including competing mortality, and no competing-risks model was
591 fitted. No clinical test is proposed here.

592 The remaining limits concern the adjustment, the biology of the metric, the single-cell evidence
593 and the pan-cancer design. Adjustment was per-gene linear residualisation on three STAR
594 metrics. RUV-III gave the same conclusions with different module memberships, so module
595 identities should not be over-read, and the residual correlation of turquoise shows what a linear
596 adjustment can leave. Biospecimen and batch variables exist for TCGA-KIRC only. In clear-cell
597 renal cancer SETD2 loss causes intron retention and transcriptional read-through^{38,39}, which
598 would raise the metric biologically. SETD2-mutant tumours had a median metric 0.47 points
599 above wild-type (46 of 368, p 0.54), small beside the plate differences, and the positional
600 gradient appears in cancer types where SETD2 loss is rare. A tumour-intrinsic component in
601 renal cancer remains possible. The metric indexes non-exonic read content and not degradation,
602 which in poly(A) libraries lowers the intronic share⁴⁰. Gene-level union counting, as in the GDC
603 pipeline, under-quantifies overlapping and antisense lncRNAs⁴¹, and the controlled-access reads
604 needed to requantify them were not used. Single-cell localisation covers 18% to 44% of lncRNA
605 module members against normal epithelium, rests on donor-level inference in three or four
606 cohorts, one with malignant labels transferred from another, and shows where a programme is
607 expressed, not whether it drives outcome. The single-cell test of the correction covers the
608 lncRNAs detectable in single cells and was run in one bulk cohort. Function was inferred by
609 guilt-by-association, and because the two networks were built separately one programme can
610 appear in both, as blue and green do. The pan-cancer analysis applied the discovery network
611 parameters to every cancer type without tuning and used one overall-survival endpoint, with
612 stage where recorded, and its dose-response analysis was specified after the locked rules had
613 been applied. Current bulk transcriptomic foundation models cover few or no lncRNAs^{42,43} and
614 cannot yet stand in for these modules. Finally, the sex-disaggregated estimates are descriptive,

both cohorts are convenience samples of consented research participants, race and ethnicity were incompletely recorded and not modelled, and the study was not registered.

Four recommendations follow for network and signature studies of bulk lncRNA expression. First, report per-sample alignment metrics and their within-cohort spread, and test whether the leading axes of the expression matrix and the eigengene of its largest module track them, preferably with a transcript-integrity measure computed from the sequencing files⁴⁴. Second, where they do, build the network on adjusted expression and report module-level associations with and without adjustment. Third, do not treat module preservation as evidence that a module is biological, and check cellular localisation or another readout independent of library preparation. Fourth, benchmark prognostic claims against full TNM with grade, cross-validate feature definition, and validate across library protocols, stating the protocol difference.

A routine alignment metric carries the leading axis of lncRNA expression in these renal cancer cohorts and, to a varying degree, across TCGA. It sets the structure of the lncRNA co-expression network and is itself structured by processing batch, and correcting for it does not detectably remove the tumour biology that single cells report. Adjustment changes the network and which modules look prognostic, and it shifts the hazard ratios of published signatures without abolishing any. The surviving programmes replicate in direction, transfer across protocols and localise to specific cell types: proximal-tubule differentiation, proliferation, and a malignant-cell lncRNA programme coupled to both. None of this makes lncRNA expression uninformative in renal cancer. It means that a network or prognostic claim made from it can be assessed only when the alignment metrics are reported alongside it.

Methods

Cohorts and data sources

Four renal cancer cohorts were quantified by one Genomic Data Commons STAR-Counts pipeline^{45,46}, giving comparable gene identifiers and FPKM definitions without cross-platform rescaling. TCGA-KIRC was the discovery cohort³⁵, CPTAC-3 the external validation cohort^{47,48}, and TCGA-KIRP and TCGA-KICH the subtype comparators. TCGA libraries are poly(A)-selected, whereas CPTAC-3 libraries are ribo-depleted stranded total RNA. All libraries came from primary tumour resection specimens. Systemic and adjuvant therapy is not recorded in the harmonised fields and was not modelled, so an unrecorded fraction of each cohort will have received systemic therapy after nephrectomy. TCGA-KIRC tumours came from 20 tissue source sites, and CPTAC-3 contributing institutions are not identified. Grade is Fuhrman nuclear grade in both cohorts. Grade and pathological stage were assigned by the contributing institutions' pathologists and not re-reviewed.

Expression files, manifests, sample types and harmonised clinical fields came from Genomic Data Commons Data Release 46.0 (10 August 2026). TCGA-KIRC grade came from the Biospecimen Core Resource clinical XML supplements, somatic calls from the open-access masked MAF files of the aliquot-ensemble whole-exome workflow, and gene annotation from GENCODE v36⁴⁹. STAR-Counts are gene-level, so every analysed feature is a gene. Four TCGA-KIRC barcodes carry two files, so expression, library size and STAR metrics were selected by file identifier.

Sample inclusion and clinical variables

Only primary tumours were analysed. A library failed if assigned reads were below 10 million or the non-feature read fraction exceeded 30%. The non-feature threshold is an outlier rule within

one library protocol and was applied to the poly(A) cohorts only, because under ribo-depleted total RNA the whole distribution of the metric lies above 30%. No CPTAC-3 library failed on read depth, and five TCGA-KIRC libraries failed. The rule was applied per file, before selecting one sample per patient, so a patient with a second usable aliquot kept it. Where a patient still had several aliquots, the one with the most assigned reads was retained.

Overall survival was days to death for deceased patients. For censored patients it was the longest follow-up across the Genomic Data Commons diagnosis and follow-up records and the Biospecimen Core Resource follow-up records, because the harmonised field alone is missing for most living TCGA-KIRC patients. A death recorded in either source counted as an event. Follow-up was administratively censored at ten years, reached by 13 discovery patients. Median follow-up used reverse Kaplan-Meier. Patients with non-positive or missing survival time were excluded.

Stage was parsed from the roman numeral of the pathological stage field, ignoring the sub-stage letter, and T category from the pathological T field, never imputed from stage. NX and MX were coded N0 and M0 and flagged, following the SSIGN and Leibovich convention of assigning pNX to pN0, because lymphadenectomy is not routine in this disease. Patients with AJCC stage IV, a T category below T4 and no explicit M category were reconciled to M1 (2 TCGA-KIRC and 12 CPTAC-3 patients), and an explicit M0 was never overridden. Grades GX and not reported were treated as missing. Sex came from the harmonised clinical record, with female as reference. No primary analysis was disaggregated by sex, and the sex analyses (Additional analyses) are sensitivity analyses. Race and ethnicity, incompletely recorded in both programmes, were not modelled, and no fairness assessment was performed.

CPTAC-3 cases were retained with a primary diagnosis of clear cell or renal cell carcinoma, not otherwise specified, because the programme codes its clear-cell cohort under the latter term. The harmonised fields carry no accrual dates, so no accrual window is stated, and the Genomic Data Commons snapshot of 10 August 2026 is the effective end of follow-up.

Ethics

The TCGA and CPTAC programmes obtained written informed consent from every participant under the approvals of their contributing institutions. This secondary analysis of de-identified, publicly available data required no further ethical approval. No patients or members of the public were involved in the design, conduct or reporting of the study.

Expression processing and adjustment for the alignment metrics

Analyses used $\log_2(\text{FPKM} + 1)$ of the Genomic Data Commons unstranded FPKM values, whose denominator is the number of reads mapped to protein-coding genes. More unprocessed RNA in a library raises both the non-feature fraction and the intronic reads counted to overlapping lncRNA loci, inflating lncRNA FPKM relative to protein-coding FPKM. The unstranded column was used in every cohort so that one quantity is compared throughout. For the stranded CPTAC-3 libraries it sums both strands, raising apparent antisense lncRNA expression relative to a strand-aware count. Protein-coding genes were retained with FPKM of at least 1.0 in at least 25% of samples, and lncRNAs with FPKM of at least 0.30 in at least 20%. The most variable genes by median absolute deviation were then kept, up to 12,000 protein-coding genes and 6,000 lncRNAs. The protein-coding cap was reached (12,000 of 19,962 genes), and 3,442 of the 16,889 unique GENCODE v36 lncRNA genes passed the abundance filter. Duplicate gene symbols kept the most variable Ensembl entry.

The four STAR summary rows (unmapped, multimapping, non-feature and ambiguous reads) were expressed for every library as percentages of the sum of those four counts and the reads assigned to annotated genes. The non-feature read fraction, the principal metric, is therefore the percentage of all reads in the STAR gene-counts summary assigned to no annotated feature. With the multimapping fraction and log10 assigned reads, it forms the three alignment metrics. Adjustment for the alignment metrics was applied to expression before network construction. Each gene was regressed by ordinary least squares on the three standardised metrics, and its residuals were kept with the gene mean added back. The stored centring, scaling and coefficients allow held-out samples to be adjusted with training-fold coefficients. The adjustment was fitted within each cohort and never transported between cohorts, because the CPTAC-3 non-feature fraction lies far outside the TCGA range. All analyses downstream of network construction use the adjusted matrix unless the observed matrix is named. Tumour composition was not removed from expression, because stromal and immune content are prognostic in this disease. ESTIMATE stromal and immune scores from the full protein-coding matrix were instead entered as covariates⁵⁰. The illumina platform setting does not return the Affymetrix-calibrated purity value, so the ESTIMATE score is an inverse-purity proxy.

Biospecimen, plate and batch variables

For TCGA-KIRC, RNA integrity number, A260/A280 ratio, slide percent necrosis and slide percentages of tumour nuclei, stromal cells and normal cells were taken from the Genomic Data Commons biospecimen records, averaging the integrity number over the RNA analytes of the primary-tumour sample. One integrity number recorded as zero is a coded missing value, excluded from Supplementary Fig. 4a, b and retained elsewhere as one of the 527 recorded values. Sequencing plate and tissue source site were fields 6 and 2 of the 28-character RNA

aliquot barcode, and Biospecimen Core Resource batch was the administrative batch number in the clinical XML. TCGA centralised extraction, library preparation and sequencing, so tissue source site is a proxy for collection, ischaemia and freezing practice. Levels with fewer than ten patients were pooled into an "other" stratum, recomputed on each analysis's complete cases. The other three cohorts lack equivalent metadata, so these analyses are restricted to TCGA-KIRC.

Network construction and module definition

Signed weighted gene co-expression networks with signed topological overlap were built separately for protein-coding genes and lncRNAs with WGCNA⁵¹, in a single block of up to 25,000 genes. Samples with standardised connectivity below -2.5 were removed as outliers, leaving 506 samples for the protein-coding network and 511 for the lncRNA network. Outlier removal preceded adjustment for the alignment metrics, followed by power selection and module detection, so module definitions do not inherit the axis. The two gene classes differ in abundance and variance scale and need separate soft thresholds, so they were kept apart. One programme can therefore appear as a module in each network, as lncRNA blue and protein-coding green do. For the protein-coding network the soft-thresholding power was the lowest reaching a signed scale-free fit R-squared of at least 0.85 over a grid of 1 to 10 and 12 to 30 in steps of 2, giving power 18 (R-squared 0.894). For the lncRNA network a parameter sweep fixed the power at 10 (R-squared 0.811), where the automatic rule would have given 12.

The sweep scored 96 configurations (three expression variants, four powers, two deep-split values, two merge heights and two minimum module-membership thresholds) with the network-construction call of the final network, without consulting survival, stage or grade. A configuration was admissible if its largest module held at most 35% of genes, at most 30% of genes were unassigned and 5 to 40 modules were produced, and 33 qualified. The eight

configurations with the smallest largest module were ranked by mean adjusted Rand index against assignments rebuilt on ten random 80% patient subsamples. The selected configuration (power 10, deep split 2, merge height 0.15, minimum module membership 0.3) had a mean adjusted Rand index of 0.605, which counts unassigned genes as one class and so inflates agreement. The sweep preceded the final pipeline run, so it documents how parameters were chosen and does not describe the analysed network.

Modules were detected with a minimum size of 30, with the study seed set immediately before the call. Excluding the grey module throughout left 29 module eigengenes. Module-trait correlations used only age, sex, stage and grade, never observation time.

Module eigengenes and projection

A module eigengene is the first principal component of the module's gene-standardised expression, sign-aligned to correlate positively with its genes' mean standardised expression and scaled to unit variance. This definition was used everywhere. The rotation was fitted once on the adjusted discovery matrix, and the stored gene means and standard deviations, loading vector and score constants were applied unchanged elsewhere. Genes and eigengenes are standardised within the cohort being scored, and every validation quantity was also computed with the discovery constants.

The leading lncRNA axis is the first principal component of the column-centred, unscaled observed lncRNA matrix of the 511 network samples, sign-aligned to per-sample mean expression. The same computation on the observed protein-coding matrix of 506 samples is the negative control. In comparisons across sample sets, the other set was centred on the discovery gene means, projected onto the discovery loadings and scaled by the discovery standard

deviation, so one fixed quantity is compared. Each set's own leading component is a different quantity and is labelled as such.

Survival analysis

Three specifications were fitted for every module, without imputation. The unadjusted specification contains the module eigengene alone. The clinical model adds age, sex, T category, nodal status, M1 and ordinal grade, and the principal specification further adds the ESTIMATE stromal and immune scores and the three alignment metrics. Residualisation removes each gene's linear dependence on the metrics by construction, so a near-zero correlation after adjustment does not show that their influence is gone. The metrics therefore also enter the principal specification as covariates, where they can absorb non-linear and score-level dependence. T category and grade were coded 1 to 4, and nodal status and M1 as binary. ESTIMATE and alignment-metric terms were standardised over all network samples. Kaplan-Meier curves use a median split and a log-rank test.

The false discovery rate was controlled by the Benjamini-Hochberg procedure⁵² at 5% within each network (13 protein-coding or 16 lncRNA modules) and is also reported over all 29 modules. Log-rank tests were adjusted in the same family, and CPTAC-3 replication across the five modules carried forward. Subtype estimates are nominal. Proportional hazards were tested by scaled Schoenfeld residuals for the eigengene term, globally for each model and for the assembled risk scores.

The leading lncRNA axis was tested in a nested sequence on one complete-case set: the axis alone, then successively adding the clinical terms, the ESTIMATE scores and the three alignment metrics, with the variance inflation factor at each step. The sequence was repeated adding the integrity number, adding the integrity number and slide necrosis, and stratifying by

sequencing plate. A parallel nested sequence took the non-feature read fraction as the exposure and ended with stratification by tissue source site. Being right-skewed, the metric was fitted on both the linear and the base-ten logarithmic scale.

Because nodal status was unassessed in half of the discovery cohort, three further specifications were fitted for every prognostic module. They coded unassessed nodal and metastatic status as separate categories, restricted analysis to patients with both assessed, and took M1 as coded without reconciliation from stage IV. SSIGN, Leibovich and UISS scores could not be reconstructed, because tumour size is absent from every TCGA-KIRC clinical XML supplement and performance status is recorded for a minority of patients. The recorded slide percent necrosis has a median of 0 and is not the whole-tumour coagulative necrosis those scores require. TNM as recorded, with ordinal grade, is therefore the strongest clinical comparator available.

Prediction models and internal validation

Penalised Cox models used an elastic-net mixing parameter⁵³ of 0.5, because a pure lasso did not converge on near-collinear hub lncRNAs. The penalty was chosen by cross-validated partial likelihood inside each training fold over a 50-value path, and comparator covariates were never penalised. The analysis set was the 493 discovery patients present in both networks with complete clinical inputs and survival (159 events).

Two feature representations were compared. The eigengene representation offers all 29 module eigengenes from the adjusted matrices, without pre-selection. The hub representation offers individual lncRNAs on observed expression, closer to what a deployable assay would measure. Its candidates were chosen inside each training fold as the 200 genes with the highest module membership in lncRNA modules reaching a 5% false discovery rate in per-module Cox models on the training rows.

Discrimination is Harrell's concordance index⁵⁴ on out-of-fold linear predictors from 10 repeats of 10-fold cross-validation. Three interval types are reported and are not interchangeable. A paired patient bootstrap of 2,000 resamples on the repeat-averaged out-of-fold predictor gives a confidence interval and p value. The 2.5th and 97.5th percentiles across the ten repeats measure cross-validation noise, and the minimum and maximum across fold assignments give a range. Apparent concordance is labelled in-sample. Module definition was also cross-validated. In three-fold cross-validation over five random fold assignments, each training fold was residualised on its own alignment metrics, both networks were rebuilt and loadings fitted on it, and held-out patients were scored with that fold's centring, scaling and rotation. The comparator was cross-validated on the same assignments, and a matched fixed-module arm used identical folds and inner splits. No formal sample-size calculation was performed, and all eligible patients were used. The bootstrap interval for the cross-validated increment has a half-width of 0.017, implying a standard error of about 0.008, so the smallest concordance increment the design could separate from zero at 80% power is about 0.024. CPTAC-3 carries 32 events, well below the 100 to 200 recommended for external validation of a survival model³⁷.

Association replication and predictive validation

Two validation analyses were run in CPTAC-3 and reported separately, because they hold different quantities fixed. The association replication fixed module membership and eigengene loadings in discovery, projected CPTAC-3 expression onto those loadings without recomputing components, and re-estimated only the Cox coefficient. Two covariate sets were fitted and both are reported. The twelve-parameter principal specification from discovery gives 2.7 events per variable on 32 events, below the threshold of five applied to the subtype cohorts. A parsimonious

six-parameter set (eigengene, age, sex, T category, M1 and ordinal grade) gives 5.3 and meets that threshold, so replication is claimed only where it holds in this arm. Directional agreement across the five modules was summarised by a one-sided sign test. This analysis tests an association, and the predictive validation tests a fixed prediction rule.

The predictive validation also fixed the penalised coefficients in discovery and applied the linear predictor unchanged to CPTAC-3. Absolute risks use a Breslow baseline hazard estimated in discovery with the locked predictor as an offset, so validation risks are out-of-sample predictions. Calibration is summarised by the calibration slope, refitted in the cohort being scored, and by the Kaplan-Meier ratio of observed to expected risk. Brier scores use inverse-probability-of-censoring weighting⁵⁵ and carry no interval. Decision-curve analysis⁵⁶ on the discovery baseline assessed clinical utility at thresholds of 0.02 to 0.60, with net benefit summarised over the windows 0.10 to 0.40 and 0.05 to 0.30 by a paired bootstrap of 500 resamples of the median gain. No threshold is proposed as a decision rule. Three years is the primary horizon, because few CPTAC-3 patients remain at risk at five.

The alignment metrics are not comparable across library protocols. The CPTAC-3 non-feature fraction has a median of 40.5% against 4.6% in TCGA-KIRC, so a discovery-standardised value would lie many standard deviations from the discovery mean and dominate any linear predictor containing it. The expression-level adjustment and the augmented model are therefore refitted within each cohort, and both standardisation arms are reported for every validation quantity. A drop-one-module test set each retained coefficient to zero individually, then all retained lncRNA module coefficients together, and re-bootstrapped the change in concordance.

Subtype specificity and module preservation

TCGA-KIRP and TCGA-KICH primary tumours were scored through the same path as every other cohort: $\log_2(\text{FPKM} + 1)$ on the network gene sets, within-cohort adjustment for the alignment metrics, then the fixed discovery loadings with cohort standardisation. The same parsimonious adjustment set (age, sex, T category, M1 and the non-feature read fraction, six parameters per model) was used in TCGA-KIRC, TCGA-KIRP and TCGA-KICH, and only cohorts with at least five events per variable contributed to inference. TCGA-KIRC (26.7) and TCGA-KIRP (6.7) qualify and TCGA-KICH (1.5) does not, so TCGA-KICH estimates are descriptive and a null there is uninformative. Between-subtype heterogeneity was tested by Cochran's Q with I-squared across qualifying cohorts only. Module preservation was assessed with the WGCNA module preservation statistics²⁰, with the adjusted discovery matrix as reference and test networks built as in the replication and subtype analyses. Permutations numbered 100 for the lncRNA network and 30 for the protein-coding network, and test module size was capped at 1,000. A Zsummary above 10 was read as strong preservation, 2 to 10 as weak to moderate, and below 2 as none.

Functional characterisation

Gene Ontology biological-process over-representation was tested with clusterProfiler⁵⁷ against a universe of the 12,000 network protein-coding genes, with Benjamini-Hochberg adjustment and cut-offs of 0.05 on the adjusted p value and 0.10 on the q value. Because partner sets are selected by correlation with an eigengene, these p values are uncalibrated and serve only as descriptive rankings.

Because lncRNAs carry little direct annotation, prognostic lncRNA module function was inferred by guilt-by-association. Protein-coding genes were ranked by absolute correlation with the module eigengene, and over-representation was tested in the top 100, 300 and 500 partners

on both the observed and the adjusted matrix. Partners were also split by correlation sign, and each side's top 300 was tested by a separately implemented hypergeometric test with the same annotation and universe. This sign-split analysis correlated residualised lncRNA eigengenes with the observed protein-coding matrix, so its partner sets can carry variance from the alignment metrics. Coupling with the catabolic module was tested directly by correlating each prognostic lncRNA eigengene with the catabolic protein-coding eigengene in each cohort. The catabolic module is identified by its Gene Ontology enrichment, because colour labels change whenever a network is rebuilt. Named module members were screened by gene symbol against the renal cell carcinoma literature, without a systematic search. LNCSTR is NCBI Gene 109729161 (HGNC:52753), retrieved 23 August 2026.

A marker score tested whether the catabolic module reflects retained proximal-tubule differentiation. It is the mean per-gene z score of documented proximal-tubule markers on the observed matrix²¹, computed for all 21 markers, the 14 inside the analysed network, those whose Human Protein Atlas specificity names proximal tubule, and the 7 outside the network. Tumours were placed on the ccA and ccB axis²² by correlating each standardised 34-gene profile with a template of +1 for ccA genes and -1 for ccB genes²³. The published centroids are not distributed, so this score reproduces the ccA and ccB axis but not the classifier. Every module eigengene was evaluated both as adjusted and as refitted on the observed matrix, because only the latter can carry variance from the metric or from admixture.

Additional analyses

Correlates of the metric. Candidate correlates of the non-feature read fraction were tested in TCGA-KIRC: a 51-gene hypoxia metagene⁵⁸, a proliferation score, the ESTIMATE scores, the slide tumour-nuclei percentage, and the eigengenes of both networks on the observed and

residualised matrices. The proliferation score is the mean per-gene z score of $\log_2(\text{FPKM} + 1)$ over MKI67, TOP2A, PCNA, MCM2, BUB1, CCNB1, CDK1, TYMS, RRM2 and AURKA, a set defined ad hoc without an external source. Variance in the metric was partitioned between a block of annotated biology and a technical block of RNA integrity, sequencing plate and tissue source site. Any block built from observed expression recovers the metric from the matrix it was measured on, so such rows are reported only beside negative controls (eigengenes of size-matched random gene sets and the leading principal components of the same matrix).

Attenuation was assessed by adding each candidate to the clinically adjusted exposure model, and rows whose added covariate is largely the exposure itself are labelled collinearity diagnostics (Supplementary Table 9).

Matched normal libraries. The 72 TCGA-KIRC and 168 CPTAC-3 Solid Tissue Normal files were processed as the tumours were, except that only the read-depth criterion of the library-failure rule applies to CPTAC-3. The discovery axis was projected onto each set, and each set's own leading component was computed and compared with a size-matched baseline of 200 discovery subsamples. Tumour-versus-normal differences are reported unpaired, within patient, and restricted to plates occupied by both sides, because the unpaired contrast is confounded by plate (Supplementary Table 3).

Published signatures. Six published TCGA-KIRC lncRNA signatures were re-scored on the discovery tumours^{4,5,6,7,8,9}. They were chosen purposively, without systematic search, one per successive biological label applied to TCGA-KIRC lncRNA models between 2019 and 2024, and are illustrative, not representative, of that literature. Each was scored as the weighted sum of $\log_2(\text{FPKM} + 1)$ over its mapped genes, with the published weights and with coefficients re-estimated in TCGA-KIRC. Two signatures publish coefficients, three publish neither coefficients

nor a hazard direction and were signed in sample, and one mapped too few genes to report beyond a footnote. Weight provenance is given on every table row, and ranges are never pooled across provenance classes. Each score was run alone, with the clinical terms, with the clinical terms and the three metrics, and on the adjusted matrix with the clinical terms. Two nulls of 1,000 random sets per signature size were built from one set of draws. Outcome-signing orients every random set along the cohort's dominant prognostic axis, so the outcome-signed null¹¹ is the reference for hazard ratios and a sign-randomised null that for correlation with the metric (Supplementary Table 13).

Normalisation. Raw STAR counts were re-read by file identifier and checked against recorded library sizes. Five lncRNA normalisations were compared: $\log_2(\text{FPKM} + 1)$ as used, $\log_2$ counts per million by library size, $\log_2$ counts per million with trimmed-mean-of-M-values factors⁵⁹, a blind variance-stabilising transformation⁶⁰, and quantile-normalised $\log_2(\text{FPKM} + 1)$ ⁶¹. With module membership fixed, eigengenes were recomputed on the trimmed-mean scale, and the between-scale difference in log hazard ratio was bootstrapped over 2,000 paired resamples (Supplementary Table 2).

Alternative endpoints. Disease-specific survival and the progression-free interval came from the TCGA Clinical Data Resource, censored at ten years as for overall survival⁶². Overall survival from that source matched the pipeline endpoint on the event indicator for all 528 patients, with survival times agreeing at Spearman 0.998. The module increment was re-estimated for each endpoint by the same repeated cross-validation, with the inner fold split specified explicitly. The false discovery rate was controlled across the four endpoint analyses reported, counting overall survival twice (pipeline and Clinical Data Resource), and across the three distinct endpoints. The disease-specific analysis was repeated with both networks rebuilt inside every training fold,

against a matched fixed-module arm. CPTAC-3 carries neither endpoint, so these are discovery-only sensitivity analyses (Supplementary Table 8).

Driver mutations and positional classes. Non-synonymous variants in VHL, PBRM1, BAP1, SETD2, KDM5C, MTOR and TP53, and the total non-synonymous count, were tested against the non-feature read fraction and the lncRNA axis, controlling the false discovery rate within each outcome over eight tests. Masked variant files were available for 368 of the 528 discovery patients, and availability itself is associated with the exposure, so this analysis is conditional on a biased subset. The 3,442 network lncRNAs were classified against GENCODE v36 protein-coding genes as exonic sense, antisense, intronic, sense-overlapping, divergent or intergenic. Per-gene correlation with the metric was compared across classes by Kruskal-Wallis and by sequential analysis of variance entering class after mean log2 expression and log10 gene length (Supplementary Table 1).

Sex. In TCGA-KIRC, CPTAC-3 and TCGA-KIRP, the five prognostic modules were refitted with a module-by-sex interaction and separately within women and within men. The sex-specific fits let every covariate and the baseline hazard differ by sex, so the two models estimate different quantities and are never presented in one row. Every module eigengene, the three alignment metrics and the lncRNA axis were tested for association with sex in all four cohorts. X- and Y-encoded module members were counted against the GENCODE background, and recorded sex was checked against XIST and the chromosome Y lncRNAs (Supplementary Table 15).

Network reorganisation

The lncRNA network was rebuilt on the observed matrix of the same 511 libraries with every main-network parameter unchanged, including power, deep split, merge height and minimum module membership. Intramodular connectivity was computed on each network's own matrix.

The partitions were compared by the adjusted Rand index over all 3,442 genes, with unassigned genes as one class, and over genes assigned in both, and each adjusted module was traced into the unadjusted partition. Positional classes in the largest unadjusted module were compared with the network background by Fisher's exact test, with Benjamini-Hochberg control across the six classes.

Single-cell localisation of modules

Four single-cell RNA-seq cohorts of untreated primary clear-cell renal cell carcinoma, none with outcome data, were used: GSE159115²⁷, the multi-region dataset of Li and colleagues²⁸, GSE222703²⁹ and GSE207493³⁰. Cells were assigned to malignant, normal epithelial and microenvironment compartments from the authors' annotations. GSE207493 lacks deposited annotations and was labelled by reference transfer from GSE159115 using only protein-coding genes, so no lncRNA influenced the malignant call. Cells in its in-tumour epithelial cluster were individually reassigned as malignant if at least three of CA9, NDUFA4L2, ANGPTL4, VEGFA and EGLN3 were detected and neither PTPRC nor PECAM1 was. Counts were summed to donor-level pseudobulk⁶³. Malignant cells were contrasted with normal epithelium, and with pooled tumour immune, endothelial and fibroblast or stromal cells, by edgeR quasi-likelihood⁶⁴, paired by donor where at least three donors contributed both groups. A cohort entered a contrast only with enough donors on both sides for inference, giving three cohorts for the normal-epithelium contrast and four for the microenvironment contrast. These contrasts come from a separate, outcome-blind pipeline, run beforehand and not altered. That pipeline also supplies the restricted-maximum-likelihood random-effects meta-analysis of lncRNA fold changes across cohorts that gives the pooled per-gene estimates in the Results.

999 For each of the 29 modules and each cohort, the statistic was the median log2 fold change of the
1000 members tested in that cohort. A two-sided empirical p value came from 2,000 random sets of
1001 the same size, drawn from the same gene class among network genes tested in that cohort and
1002 matched on single-cell abundance by deciles of log10 mean counts per million. Cohorts were
1003 combined by an unweighted Stouffer sum of per-cohort z scores, with Benjamini-Hochberg
1004 control across the 29 modules within each contrast. The null treats genes as exchangeable
1005 although genes within a module are correlated, so combined z scores overstate precision and
1006 serve only to rank modules. Inference rests on per-cohort empirical p values and cross-cohort
1007 agreement.

1008 **Pan-cancer analysis**

1009 Open-access STAR-Counts files for every TCGA project were retrieved from Genomic Data
1010 Commons Data Release 46.0 and verified against their md5 checksums. Primary tumours
1011 (peripheral-blood cancer for TCGA-LAML) and solid-tissue normals were processed with the
1012 discovery rules: the library-failure rule, one library per patient per group, the discovery gene
1013 filters applied within each project and group, and the discovery definition of the leading
1014 component. Each project therefore uses its own expressed genes, so its own-component values
1015 differ from those computed on the discovery genes (Supplementary Table 4). TCGA-KIRC run
1016 through this pipeline is the like-for-like check. The discovery classifier assigned a positional
1017 class to every GENCODE v36 lncRNA and reproduced the discovery class of all 3,442 network
1018 lncRNAs. The lncRNA network was built twice per eligible project, on observed and
1019 residualised expression, with the discovery lncRNA parameters and sample-outlier rule. Overall
1020 survival came from the TCGA Clinical Data Resource, censored at ten years. Stage was the
1021 AJCC pathologic stage where at least 70% of analysed patients had it, otherwise the clinical

stage, and was otherwise omitted. Project estimates were combined by restricted-maximum-likelihood random-effects meta-analysis⁶⁵. Decision rules were written into the pipeline configuration, committed to version control and hashed before any non-renal pan-cancer project was analysed. The renal projects had been analysed during discovery, so the thresholds were set with those results known. Pipeline stages 39 to 41 verify the rules before running, and the rule choosing the source of tumour stage for survival models is fixed in the analysis script. A project was eligible with at least 100 retained primary tumours, and TCGA-LAML, a blood cancer, was excluded from inference. A lncRNA-specific axis required the lncRNA component to track the non-feature fraction at an absolute Spearman rho of at least 0.5 and to exceed the protein-coding component by at least 0.3. A reorganised network required a largest observed module holding at least 40% of genes, an eigengene correlating with the metric at an absolute Pearson r of at least 0.7, and an adjusted Rand index below 0.2. Normal-tissue sets with at least 20 libraries were analysed descriptively, and projects with at least 20 deaths entered the meta-analysis. The dose-response analysis, specified after these rules were applied, correlated four measures of the metric's within-cohort spread with the rule quantities and the plate share of the metric's variance across projects, with Benjamini-Hochberg control across its 20 tests.

Single-cell test of the correction

The test used the TCGA-KIRC primary tumours and solid-tissue normals of the pan-cancer build (532 tumours), 4,088 lncRNAs expressed in either group, and the single-cell results of the outcome-blind pipeline, whose candidate lncRNAs were locked before any bulk step. The pan-cancer build retains the 4 patients excluded from discovery for survival time. The correction was the one used throughout, fitted in the tumours and applied unchanged to the normals. A

1045 sensitivity arm fitted it on both groups with a tumour indicator and removed only the metric
1046 terms. Agreement was the Spearman correlation across lncRNAs between the random-effects
1047 single-cell fold change of malignant against normal epithelial cells and the mean paired bulk
1048 tumour-minus-normal difference, together with sign agreement among lncRNAs at a single-cell
1049 false discovery rate below 0.05. The change in agreement was assessed by 2,000 gene bootstraps.
1050 Agreement was computed for the paired contrast and for the unpaired contrast of all tumours
1051 against all normals, which was added post hoc because the paired contrast proved insensitive.
1052 Scores were means of per-gene z scores. The malignant-marker score used CA9, NDUFA4L2,
1053 ANGPTL4, VEGFA and EGLN3, the markers the single-cell pipeline used to check its
1054 malignant labels. The null was 1,000 sets of lncRNAs the single-cell analysis calls non-specific
1055 (absolute log2 fold change below 0.25, false discovery rate above 0.5), matched to the tumour-
1056 specific set on bulk abundance decile. The expected direction of every test was recorded in the
1057 analysis script before its first run.

1058 **Comparison with RUV-III**

1059 RUV-III¹⁶ was applied with the ruv package to log2(FPKM + 1) of the 12,000 discovery protein-
1060 coding genes and the expressed lncRNAs of the 528 TCGA-KIRC tumours with a ccA/ccB class,
1061 following the pseudo-replicate design of RUVprps⁶⁶. Biology groups were ccA/ccB class crossed
1062 with stage I-II or III-IV. Pseudo-samples were means of at least three samples of one group on
1063 one plate, and of the ten samples with the lowest and the ten with the highest non-feature fraction
1064 or depth within a group, giving 12 replicate sets of 66 pseudo-samples. Negative controls were
1065 the 1,000 genes most associated with the metric and depth among the half least associated with
1066 the biology groups. Three unwanted factors were removed, with one and five as sensitivity
1067 analyses. The single-cell coherence test was repeated on these 528 tumours with 500 null sets, so

1068 its excess values differ slightly from those of the single-cell test of the correction. This
1069 comparison is a post hoc sensitivity analysis.

1070 **Statistics and reproducibility**

1071 All tests were two-sided with alpha 0.05. The false discovery rate was controlled by the
1072 Benjamini-Hochberg procedure at 5% within the stated families, and every reported false
1073 discovery rate names its family. Hazard ratios are from Cox proportional-hazards models, per
1074 standard deviation of the eigengene or metric, with 95% confidence intervals. Correlations are
1075 Spearman unless stated otherwise. Confidence intervals on a concordance index are the estimate
1076 plus or minus 1.96 standard errors. The index depends on the cohort's censoring distribution, so
1077 values are not comparable between cohorts and the within-cohort increment is the quantity
1078 interpreted. Bootstraps used 2,000 patient-level paired resamples (500 for the decision-curve
1079 medians), with the p value as twice the smaller tail proportion. Every model uses the complete
1080 cases of its own covariates, and its sample size and event count are given where it first appears.
1081 Randomisation and blinding do not apply to an analysis of existing cohorts. The global seed was
1082 20260816, reset at the start of every stochastic stage, and the five fold assignments of the rebuild
1083 analyses used seeds 20260817 to 20260821.

1084 Analyses used R 4.4.2 on Windows with the numeric locale set to C. The principal packages
1085 were WGCNA 1.73, survival 3.7-0, glmnet 4.1-10, data.table 1.16.4, clusterProfiler 4.12.6,
1086 estimate 1.0.13, edgeR 4.2.2, DESeq2 1.44.0, limma 3.60.6, GenomicRanges 1.56.2, metafor
1087 4.8.0 and ruv 0.9.7.2. The full list with versions is in the repository. The study was not registered
1088 and had no separate protocol, so analyses are described as principal or sensitivity analyses, not as
1089 pre-specified. The network parameters, the seed and every analysis setting are fixed in the

pipeline configuration file. Reporting follows the TRIPOD+AI statement⁶⁷, and the completed checklist accompanies this article.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

No new data were generated for this study. All primary data analysed are publicly available.

Expression data are the `augmented_star_gene_counts.tsv` STAR-Counts files of the National Cancer Institute Genomic Data Commons (<https://portal.gdc.cancer.gov>), Data Release 46.0 of 10 August 2026, for projects TCGA-KIRC, TCGA-KIRP, TCGA-KICH and CPTAC-3. Harmonised clinical fields, sample types and file manifests were retrieved from the Genomic Data Commons application programming interface at the same release. Tumour grade for TCGA-KIRC was taken from the Biospecimen Core Resource clinical XML supplements. The biospecimen measurements, which the Genomic Data Commons biospecimen records carry for TCGA-KIRC only, were the RNA integrity number, the A260/A280 ratio, the slide percentages of tumour nuclei, stroma, normal tissue and necrosis, the sequencing plate and the Biospecimen Core Resource batch. Somatic variant calls are the open-access masked MAF files of the same release. Disease-specific survival and the progression-free interval are from the TCGA Clinical Data Resource⁶². Gene annotation is GENCODE v36⁴⁹, the reference of the Genomic Data Commons STAR-Counts workflow. Single-cell data are the Gene Expression Omnibus series GSE159115, GSE222703 and GSE207493 and the dataset of Li and colleagues at Mendeley Data (g67bkbnhhg), all open access. The pan-cancer analysis used the STAR-Counts files of all

1112 33 TCGA projects at the same release. Supplementary Data 5 lists every file with its identifier,
1113 md5 checksum and retention status.

1114 The expression, clinical and mutation files are open access and require no authorisation. The
1115 parent studies are registered in dbGaP under accessions phs000178 (The Cancer Genome Atlas)
1116 and phs001287 (Clinical Proteomic Tumor Analysis Consortium). These accessions govern the
1117 controlled-access data of the same programmes, which were not used.

1118 Supplementary Data 1 lists all 1,549 STAR-Counts expression files of the four projects, one row
1119 per file, with the Genomic Data Commons file and case identifiers, workflow version, data
1120 release, md5 checksum, sample type, and whether the file was retained or why it was excluded.

1121 Local copies of 1,491 files matched the Genomic Data Commons md5 checksums. The
1122 remaining 58 files, 32 TCGA-KIRP and 25 TCGA-KICH normal-tissue libraries and one TCGA-
1123 KIRP additional-new-primary sample, were never downloaded or analysed. Supplementary Data
1124 2 assigns the 10,407 genes of the two networks to all 29 modules and gives per-gene module
1125 membership, eigengene loading, and the centring and scaling constants that define each module
1126 eigengene. Supplementary Data 3 gives the survival results for all 29 modules under the
1127 principal specification, and Supplementary Data 4 the Gene Ontology enrichment of the protein-
1128 coding modules. Source data for every figure panel are the result tables in the code repository.
1129 The repository README lists them by figure and table, and each supplementary figure legend
1130 names the table behind its panels.

1131 **Code availability**

1132 The complete analysis pipeline is available at <https://github.com/GehadYoussef/rcc-lncrna->
1133 networks and archived at [ZENODO DOI], at commit [COMMIT HASH], under an MIT licence.

1134 The pipeline is written in R. The bulk pipeline in `src/R` comprises 48 files, namely the
1135 configuration file `00_config.R`, 43 numbered stage scripts, the enrichment helper
1136 `ora_hyergeometric.R`, the pan-cancer helper `pan_helpers.R`, the download script
1137 `00_download_gdc.R` and the driver `run_all.R`. The configuration file holds every analysis
1138 constant, threshold and random seed and the locked pan-cancer decision rules. The driver runs
1139 the stages in order and caches each stage's output. The single-cell pipeline that produces the
1140 localisation contrasts is in `src/singlecell`, with its settings in
1141 `src/config/singlecell.yml`. The repository README lists the stage scripts and names
1142 the result tables behind every figure and table. The main figures are built by `13_figures.R`, the
1143 supplementary figures by `32_supplementary_figures.R` and the manuscript tables by
1144 `14_tables.R`, all directly from the result tables, so that no table or figure can drift from the
1145 number in the text.

1146 Every result table cited in the paper is in the `results` folder of the repository. Package versions
1147 are listed in the Methods (Statistics and reproducibility) and in
1148 `results/31_package_versions.tsv`, and each stage writes its session information when
1149 run.

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Author contributions

G.Y. and N.H. conceived the study. G.Y. designed the analysis, wrote the pipeline, assembled the cohorts, performed the statistical analysis, produced the figures and tables, and wrote the first draft. N.H. supervised the work, obtained the funding, administered the project, and reviewed and edited the manuscript. Both authors interpreted the results and read and approved the submitted manuscript.

Competing interests

N.H. is the co-founder and Chief Technology Officer of CardiaTec Bio, a company developing therapeutics for cardiovascular diseases, and the co-founder of KURE.ai, which focuses on AI-driven oncology drug discovery. N.H. also serves on the Scientific Advisory Board of the Institute of Cancer Research (ICR). These affiliations are unrelated to the subject matter of this manuscript. G.Y. declares no competing interests. Neither author declares any non-financial competing interests.

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1343 **Figure legends**

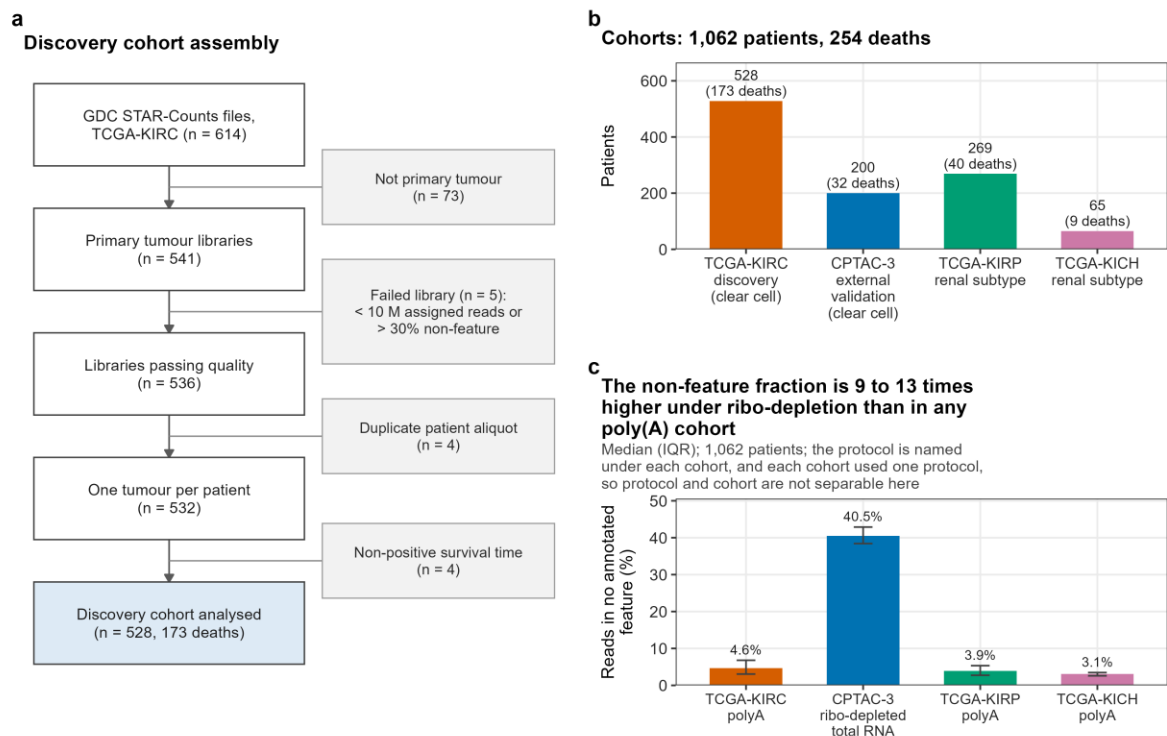


Fig. 1 | Study design, cohorts and library protocol. **a** Assembly of the discovery cohort. Of 614 TCGA-KIRC STAR-Counts files downloaded from the Genomic Data Commons, 73 were not from primary tumours, leaving 541 primary-tumour libraries. Five failed the library rule (fewer than 10 million assigned reads or more than 30% non-feature reads), leaving 536. Four were duplicate aliquots, leaving 532 tumours from distinct patients, and 4 patients had non-positive or missing survival time. The shaded box is the analysed discovery cohort of 528 patients with 173 deaths. **b** Patients per cohort, with deaths in parentheses: TCGA-KIRC 528 (173), CPTAC-3 200 (32), TCGA-KIRP 269 (40) and TCGA-KICH 65 (9), totalling 1,062 patients and 254 deaths. The TCGA-KIRP bar counts the 40 deaths among the 263 patients with a recorded T category, who enter the adjusted models. Of all 269 TCGA-KIRP patients with module scores, 42 died. Colours identify cohorts, as in **c**. **c** Median non-feature read fraction per cohort, the percentage of reads in the STAR gene-counts summary assigned to no annotated feature. The denominator sums the unmapped, multimapping, non-feature, ambiguous and gene-

1357 assigned counts. Bars are medians and whiskers interquartile ranges: TCGA-KIRC 4.6% (3.1 to
1358 6.8), CPTAC-3 40.5% (38.4 to 42.9), TCGA-KIRP 3.9% (2.7 to 5.3) and TCGA-KICH 3.1%
1359 (2.6 to 3.5), on 528, 200, 269 and 65 libraries. Library protocols are given under the cohort
1360 labels. The ribo-depleted total-RNA cohort is about 9 to 13 times higher than the poly(A) cohorts
1361 (8.7-fold against TCGA-KIRC and 13.0-fold against TCGA-KICH). Each cohort used one
1362 protocol, so protocol and cohort cannot be separated.

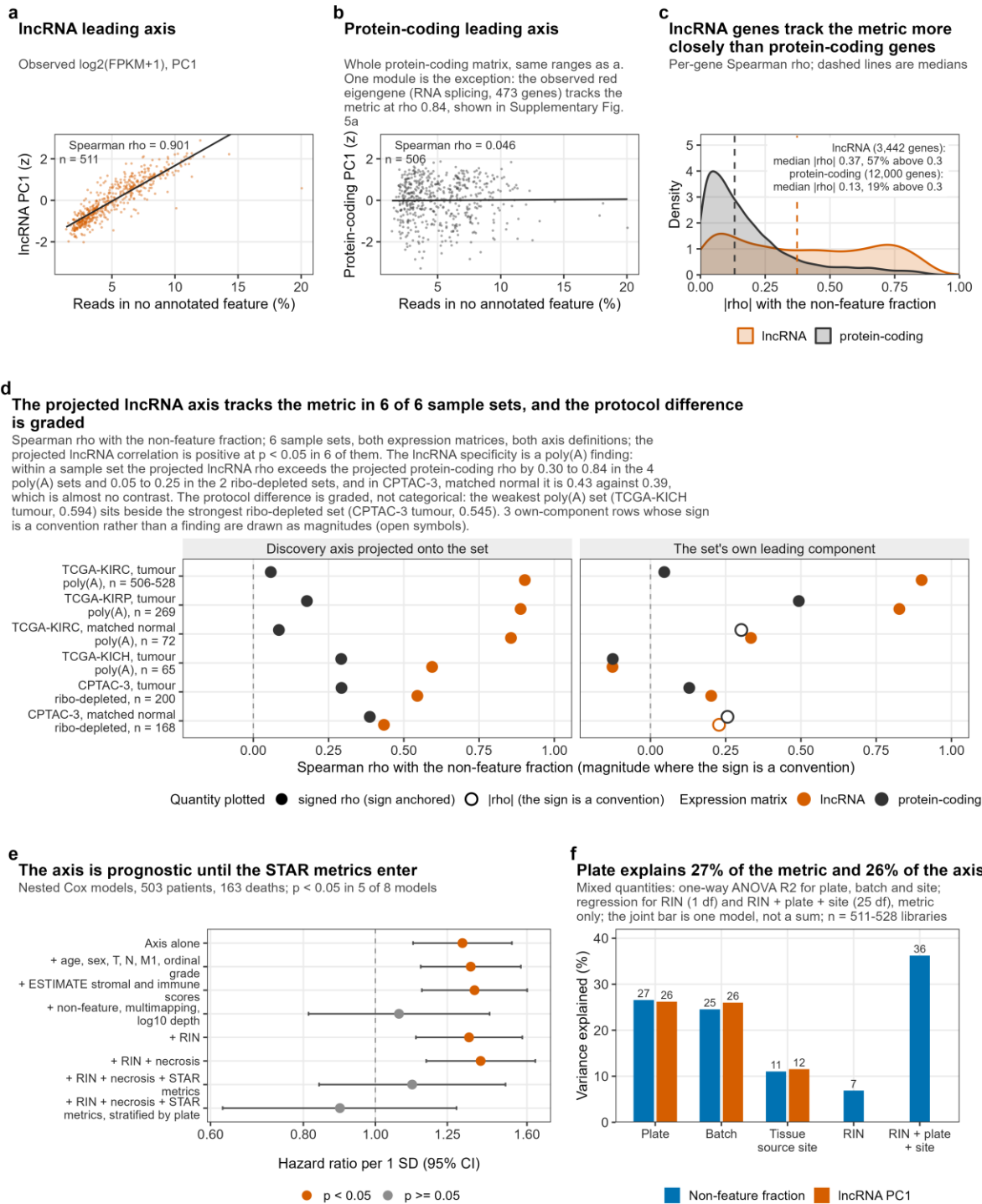


Fig. 2 | The leading IncRNA expression axis tracks non-exonic read content. **a** First principal component of the observed IncRNA matrix, log₂(FPKM + 1), against the non-feature read fraction per library, with a least-squares fit (Spearman rho 0.90, n = 511). **b** As **a** for the protein-

1366 coding matrix, on identical axes ($\rho = 0.05$, $n = 506$). The observed protein-coding red eigengene
 1367 (473 genes, RNA splicing) does track the metric ($\rho = 0.84$). **c** Densities of per-gene absolute
 1368 Spearman correlation with the metric. Dashed medians are 0.37 for 3,442 lncRNAs (56.8%
 1369 above 0.3) and 0.13 for 12,000 protein-coding genes (18.8%). **d** Spearman correlation of the axis
 1370 with the metric in six sample sets, faceted by the projected discovery axis and each set's own
 1371 leading component. Open symbols, plotted as magnitudes, mark the 3 of 12 own-component
 1372 rows whose sign cannot be fixed because its correlations with mean expression and the projected
 1373 axis are weak or disagree. The projected lncRNA axis gives $\rho = 0.90, 0.89, 0.86, 0.59, 0.55$ and
 1374 0.43 in TCGA-KIRC tumours ($n = 528$), TCGA-KIRP (269), TCGA-KIRC normal kidney (72),
 1375 TCGA-KICH (65), CPTAC-3 tumours (200) and CPTAC-3 normal tissue (168). **e** Hazard ratio
 1376 per standard deviation of the lncRNA axis under eight nested Cox models (503 patients, 163
 1377 events), with 95% confidence intervals on a logarithmic axis. Red marks p below 0.05. The
 1378 hazard ratio is 1.31 to 1.39 before the three STAR metrics enter and 1.08 after. It is 1.12 with
 1379 RNA integrity and slide necrosis added to the STAR metrics, and 0.90 with plate stratification. **f**
 1380 Variance in the metric and the axis explained by plate (26.6% and 26.2%), batch (24.6% and
 1381 26.0%) and tissue source site (11.1% and 11.6%), as one-way analysis of variance R-squared on
 1382 511 to 528 libraries. Two metric-only bars give regression R-squared on the 527 patients with an
 1383 integrity number, for RNA integrity alone (6.9%) and with plate and site (36.2%).

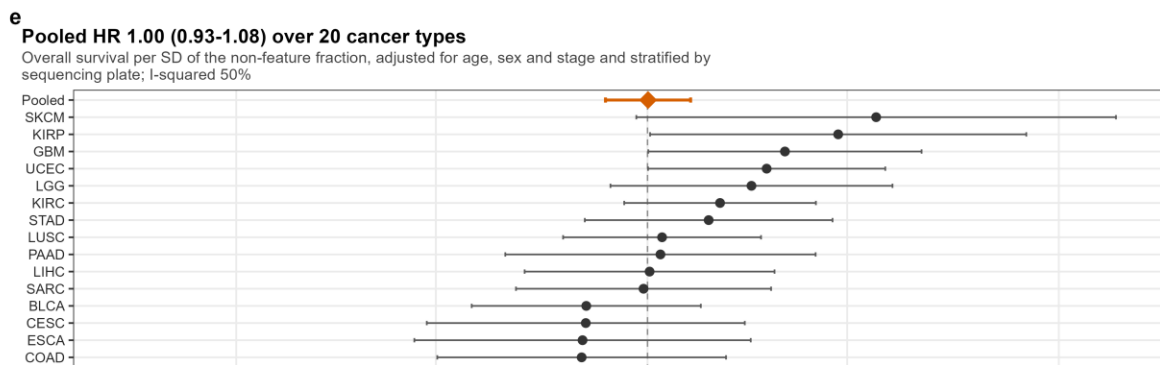
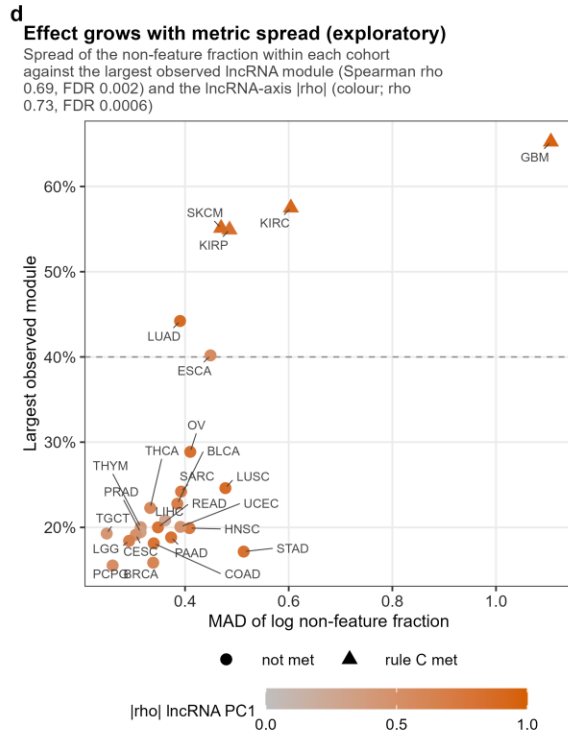
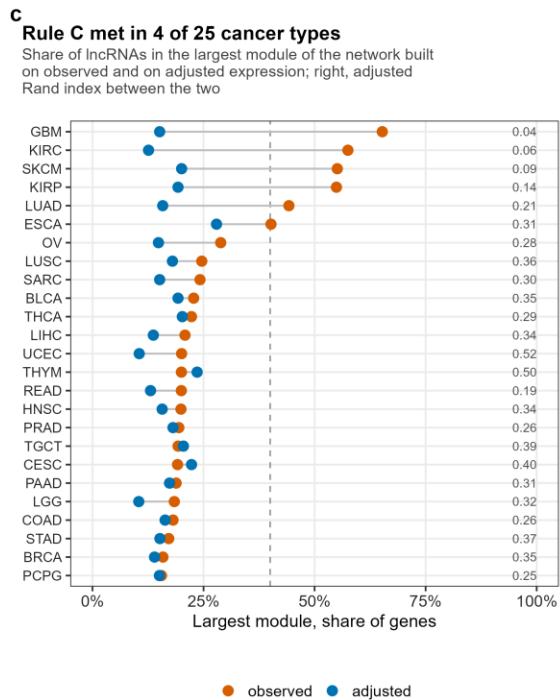
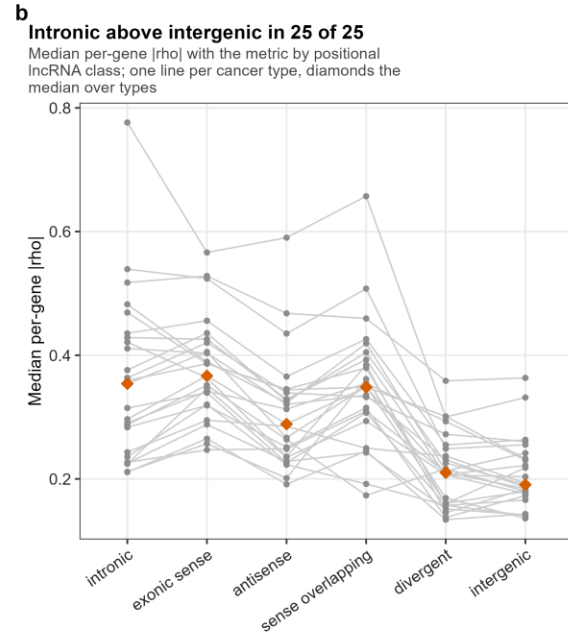
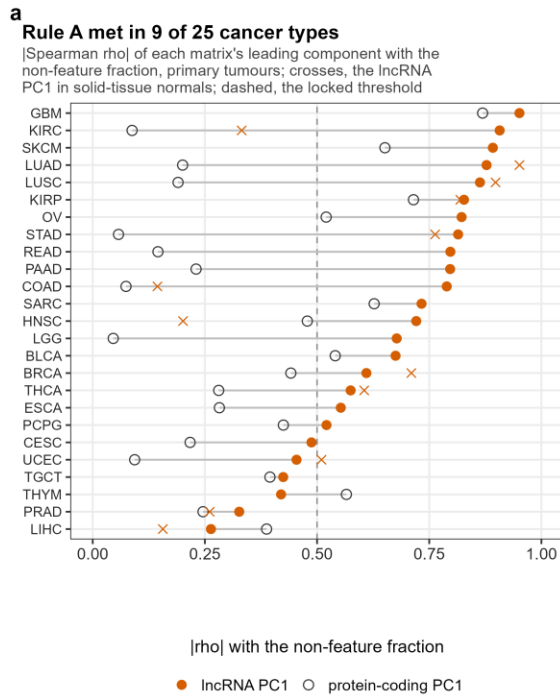


Fig. 3 | The lncRNA axis and network reorganisation across TCGA. Twenty-five cancer types with at least 100 retained primary tumours. Dashed lines are the decision rules fixed before analysis. **a** Absolute Spearman correlation with the non-feature read fraction of the leading principal component of the lncRNA (filled) and protein-coding (open) matrices, and of the lncRNA component in solid-tissue normals with at least 20 libraries (crosses). The lncRNA-specific rule (absolute rho at least 0.5 and at least 0.3 above protein-coding) is met in 9 of 25. **b** Median per-gene absolute correlation with the metric by positional lncRNA class, one line per cancer type, with diamonds for the median over types. Intronic exceed intergenic lncRNAs in all 25. **c** Share of lncRNAs in the largest module of the observed (orange) and adjusted (blue) networks, with the adjusted Rand index between the two partitions at right. The network rule is met in GBM, KIRC, SKCM and KIRP. **d** Exploratory panel, specified after the locked rules were applied. Within-cohort spread of the metric (median absolute deviation of its logarithm) against the share of genes in the largest observed module, coloured by the absolute rho of the lncRNA component. Triangles meet the network rule. **e** Hazard ratio for overall survival per standard deviation of the non-feature fraction in the 20 cancer types with at least 20 deaths, adjusted for age, sex and stage where recorded and stratified by sequencing plate. Error bars are 95% confidence intervals, and the diamond is the random-effects pooled estimate, 1.00 (0.93 to 1.08). Cancer types are TCGA project codes. Source data are in Supplementary Table 6.

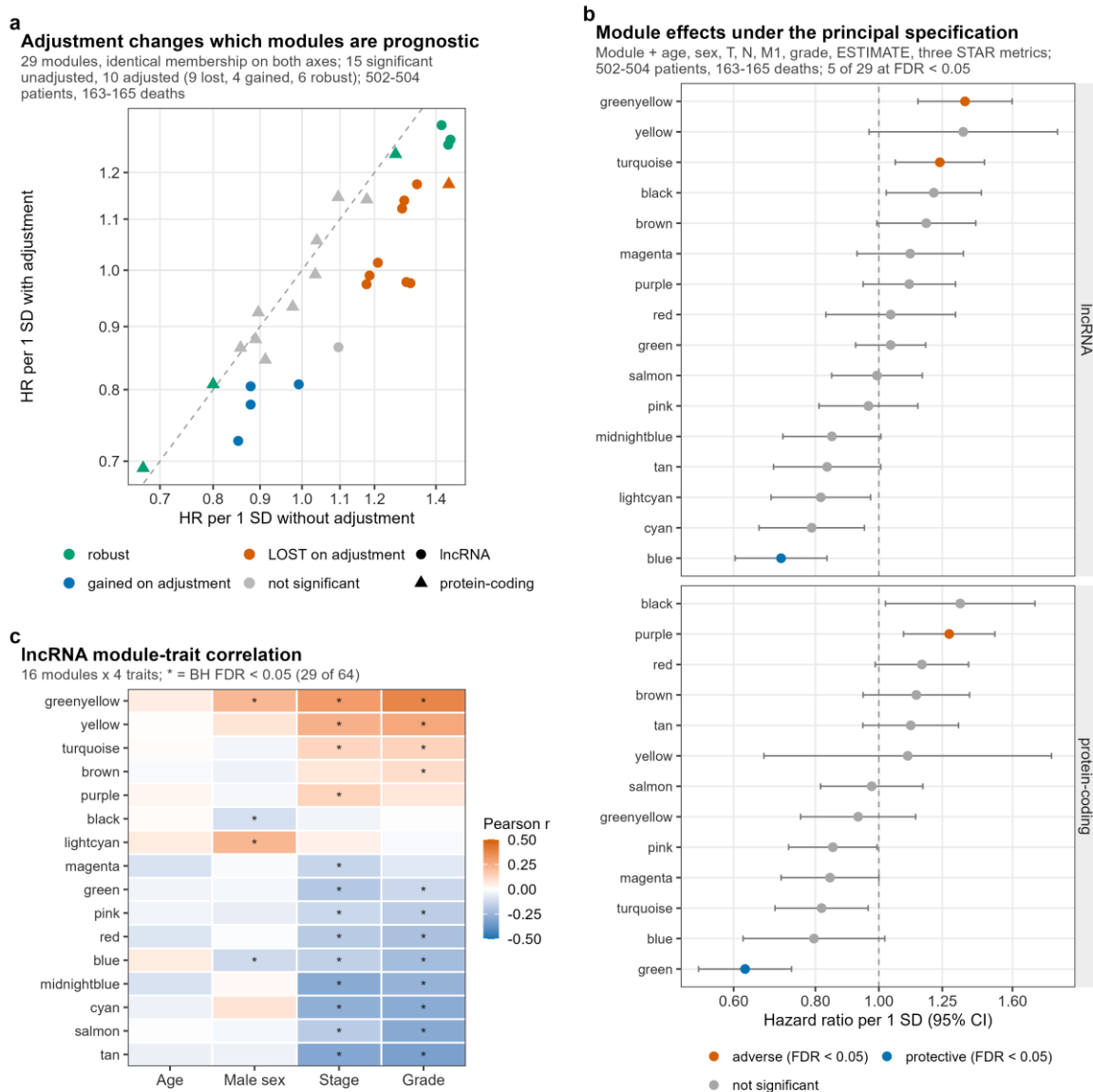


Fig. 4 | Adjusting for the alignment metrics changes which co-expression modules are

prognostic. **a** Hazard ratio per standard deviation of each of the 29 module eigengenes,

computed on the observed matrix (horizontal axis) and on the matrix residualised on the three

STAR alignment metrics (vertical axis). Module membership is the same on both axes, so only

the adjustment differs. Axes are logarithmic and the dashed line marks equality. Colour shows

the outcome of adjustment and shape the network. Of 29 modules, 15 were significant at a 5%

false discovery rate without adjustment and 10 with it: 9 lost significance, 4 gained it, 6 were

1409 significant in both arms and 10 in neither. Models adjust for age, sex, T category, nodal status,
1410 M1 and ordinal grade and are fitted on 504 patients with 163 events (lncRNA network) and 502
1411 with 165 (protein-coding network). False discovery rates are controlled within network and arm.
1412 **b** Hazard ratio per standard deviation of each module eigengene under the principal
1413 specification, by network, with 95% confidence intervals on a logarithmic axis. Colour marks the
1414 direction of modules with a within-network false discovery rate below 0.05, and grey marks non-
1415 significant modules. Five of 29 modules meet that threshold: protein-coding green and purple,
1416 and lncRNA blue, greenyellow and turquoise. **c** Pearson correlation of the 16 lncRNA module
1417 eigengenes with age, male sex, ordinal AJCC stage and ordinal grade in 511 patients. Asterisks
1418 mark a Benjamini-Hochberg false discovery rate below 0.05 across the panel's 64 tests.

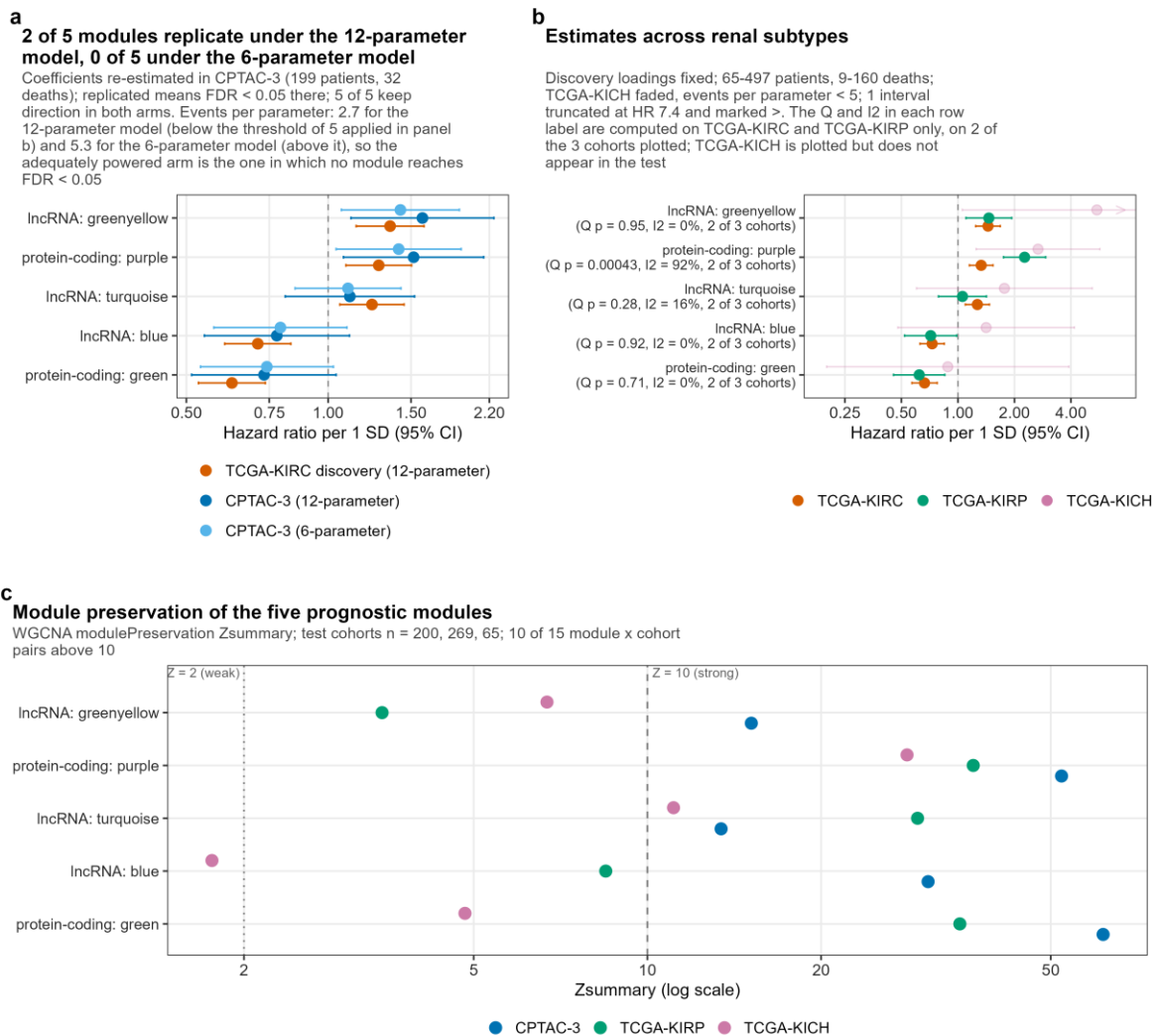


Fig. 5 | External replication, subtype estimates and module preservation. **a** Three estimates for each of the five modules significant in discovery. These are TCGA-KIRC under the principal specification (504 patients and 163 events for lncRNA modules, 502 and 165 for protein-coding modules), CPTAC-3 with the same twelve-parameter covariate set, and CPTAC-3 with a six-parameter set of module eigengene, age, sex, T category, M1 and ordinal grade. Both CPTAC-3 arms use 199 patients with 32 events, with membership and loadings fixed in discovery and only the module coefficient re-estimated. Points are hazard ratios per standard deviation with 95% confidence intervals on a logarithmic axis. All five keep the discovery direction in both arms.

1427 Greenyellow and purple are nominally significant in both. They meet a 5% false discovery rate
 1428 in the twelve-parameter arm (2.7 events per parameter, below the threshold of 5 used elsewhere)
 1429 but not in the six-parameter arm (5.3). **b** The same five modules projected with the fixed
 1430 discovery loadings onto TCGA-KIRC (497 patients, 160 events), TCGA-KIRP (263, 40) and
 1431 TCGA-KICH (65, 9), adjusted for one parsimonious set of age, sex, T category, M1 and the non-
 1432 feature fraction. Error bars are 95% confidence intervals on a logarithmic axis. Row labels give
 1433 Cochran Q p values and I-squared for TCGA-KIRC and TCGA-KIRP, the two cohorts meeting
 1434 the events-per-parameter rule. TCGA-KICH, at 1.5 events per parameter, is faded and
 1435 descriptive only. Its lncRNA greenyellow interval extends beyond the axis limit and is marked. **c**
 1436 Preservation of the five modules in CPTAC-3 (n = 200), TCGA-KIRP (269) and TCGA-KICH
 1437 (65), as WGCNA Zsummary on a logarithmic axis, from 100 permutations for the lncRNA
 1438 network and 30 for the protein-coding network. Reference lines mark the conventional weak (Z
 1439 = 2) and strong ($Z = 10$) thresholds.

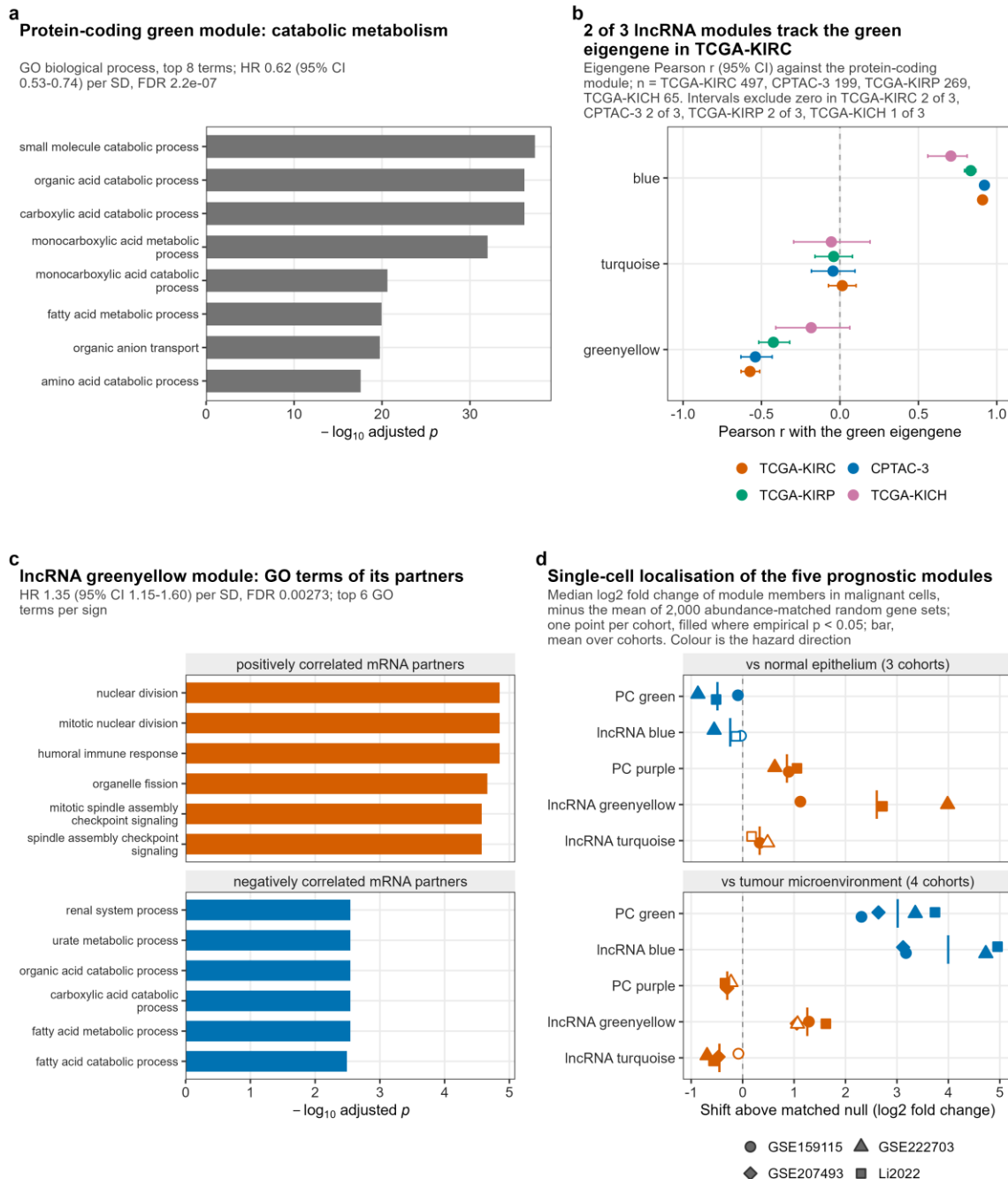


Fig. 6 | What the prognostic modules are coupled to and where they are expressed. **a** The eight Gene Ontology biological-process terms with the smallest adjusted p values for the protein-coding green module, against the 12,000 protein-coding genes entering the network. Bars are $-\log_{10}$ Benjamini-Hochberg adjusted p. Under the principal specification the module's hazard

1444 ratio is 0.62 per standard deviation (95% confidence interval 0.53 to 0.74, false discovery rate
 1445 2.2e-07). **b** Pearson correlation, with 95% confidence intervals, of each prognostic lncRNA
 1446 module eigengene with the protein-coding green eigengene in 497 TCGA-KIRC, 199 CPTAC-3,
 1447 269 TCGA-KIRP and 65 TCGA-KICH patients. Two of three intervals exclude zero in TCGA-
 1448 KIRC, CPTAC-3 and TCGA-KIRP, and one of three in TCGA-KICH. The turquoise interval
 1449 spans zero in every cohort. **c** Gene Ontology terms of the protein-coding genes whose expression
 1450 tracks the lncRNA greenyellow eigengene. The 300 most strongly correlated partners of each
 1451 sign are tested separately, and the six terms with the smallest adjusted p are shown per sign. The
 1452 module's hazard ratio is 1.35 (1.15 to 1.60, false discovery rate 0.0027). Positively correlated
 1453 partners are enriched for a proliferation and humoral-immune programme, and negatively
 1454 correlated partners for the catabolic and renal-tubular programme of **a**. **d** Single-cell localisation
 1455 of the five prognostic modules. For each module and cohort, the median log2 fold change of
 1456 members in malignant cells against normal kidney epithelium (GSE159115, GSE222703 and Li
 1457 2022) and against pooled tumour immune, endothelial and stromal cells (the same three and
 1458 GSE207493), minus the mean of 2,000 abundance-matched random gene sets of the same gene
 1459 class. Shape marks the cohort, colour the hazard direction, filled symbols an empirical p below
 1460 0.05, and vertical bars the mean over cohorts. Members tested per cohort range from 14 to 18 for
 1461 greenyellow against normal epithelium and are listed with all values in Supplementary Table 19.

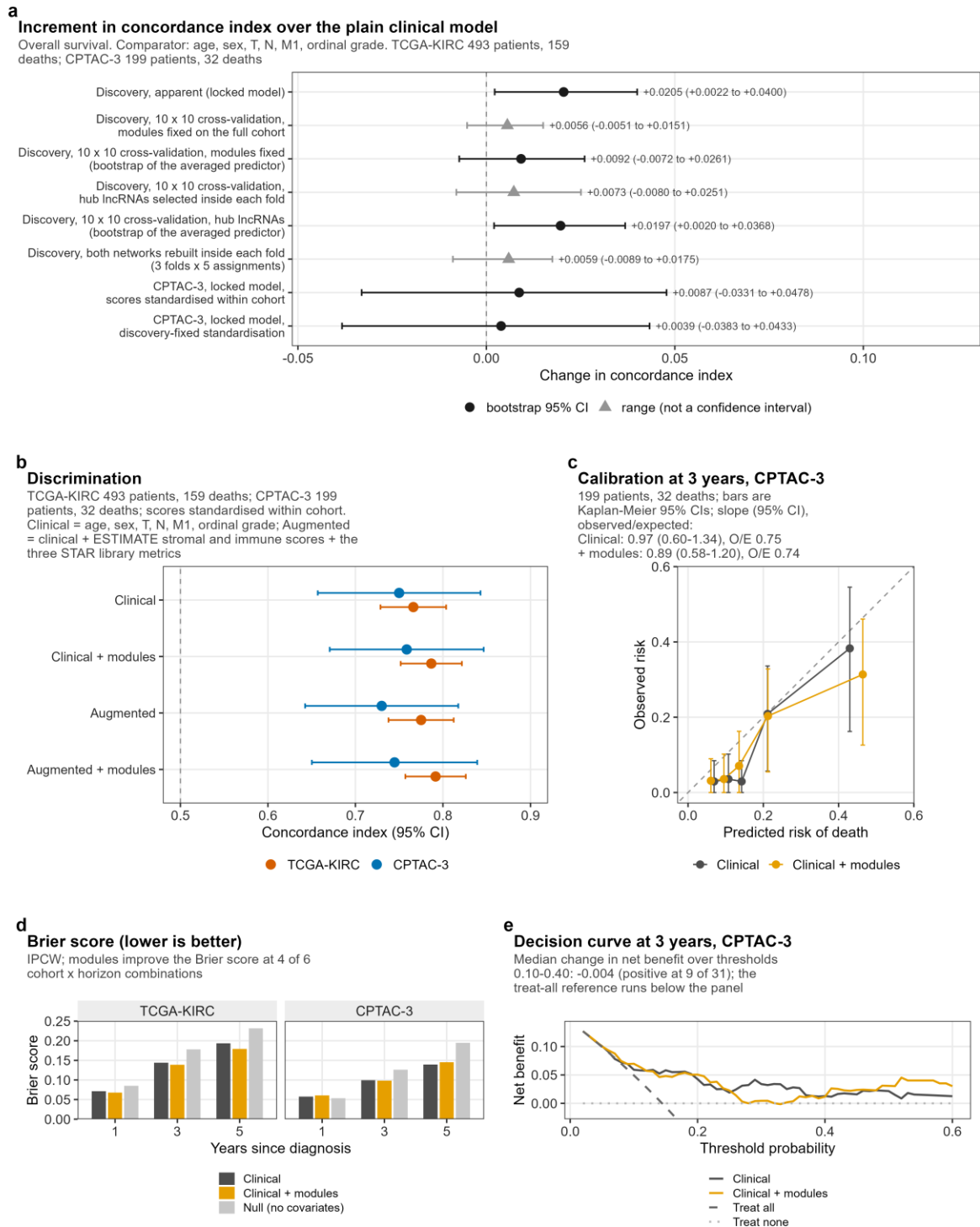


Fig. 7 | Module eigengenes add no demonstrable discrimination over the clinical model. All panels are for overall survival, with models defined in the Table 3 footnote. **a** Increment in concordance index over the clinical model under six estimation schemes (eight rows), in 493

1465 discovery patients with 159 events and 199 CPTAC-3 patients with 32. Black circles carry paired
 1466 patient bootstrap 95% confidence intervals (2,000 resamples), and grey triangles carry ranges
 1467 across repeats, which are not confidence intervals. Row one is the locked model scored in sample
 1468 (+0.0205). Rows two to five are ten repeats of ten-fold cross-validation with module eigengenes
 1469 (+0.0092) and with hub lncRNAs selected inside each fold (+0.0197), each range row preceding
 1470 its bootstrap row. Row six rebuilds both networks inside each training fold, over three folds and
 1471 five assignments (+0.0059). The last two rows are the locked model in CPTAC-3 under cohort
 1472 (+0.0087) and discovery-fixed (+0.0039) standardisation. **b** Concordance index with 95%
 1473 confidence intervals for four locked models in both cohorts, eigengenes standardised within
 1474 cohort. The dashed line marks 0.5, and TCGA-KIRC values are apparent (in-sample). **c** Three-
 1475 year calibration in CPTAC-3 by quintile of predicted risk, with the baseline hazard fixed in
 1476 discovery. The dashed line is perfect calibration, and error bars are Kaplan-Meier 95%
 1477 confidence intervals. The calibration slope is 0.97 (0.60 to 1.34) for the clinical model and 0.89
 1478 (0.58 to 1.20) with modules, and observed over expected risk is 0.75 and 0.74. **d** Brier scores at
 1479 one, three and five years, weighted by the inverse probability of censoring (lower is better), for
 1480 the clinical model, the model with modules and a null model assigning every patient the Kaplan-
 1481 Meier risk. Modules improve the score at 4 of 6 cohort-by-horizon combinations, three of them
 1482 apparent TCGA-KIRC scores, and at 1 of 3 horizons in CPTAC-3. **e** Three-year decision curve
 1483 in CPTAC-3 with the same fixed baseline, thresholds 0.02 to 0.60. The median change in net
 1484 benefit over thresholds 0.10 to 0.40 is -0.004, and the change is positive at 9 of 31 thresholds.

1485 **Tables**

1486 **Table 1 | Characteristics of the four renal cancer cohorts.**

Cohort	Patients	Deaths	Median follow-up (days)	Median age (years)	Male (%)	T3 or T4 (%)	M1 (%)	Grade 3 or 4 (%)	Non-feature reads (%), median (IQR)
TCGA-KIRC (discovery, clear cell)	528	173	1,657	61.1	65.0	36.0	15.3 (16.0)	53.8	4.6 (3.1 to 6.8)
CPTAC-3 (external validation, clear cell)	200	32	1,414	61.5	71.5	36.2	8.0 (5.7)	40.0	40.5 (38.4 to 42.9)
TCGA-KIRP (papillary)	269	40	805	62.2	72.9	23.2	5.2 (9.1)	not recorded	3.9 (2.7 to 5.3)
TCGA-KICH (chromophobe)	65	9	2,752	50.7	58.5	30.8	3.1 (5.6)	not recorded	3.1 (2.6 to 3.5)

1487 All four cohorts were quantified with the same Genomic Data Commons STAR-Counts pipeline.

1488 TCGA libraries are poly(A)-selected and CPTAC-3 libraries are ribo-depleted total RNA.

1489 Median follow-up is the reverse Kaplan-Meier median.

1490 Percentages are of patients with the variable recorded. T category is recorded for 528 of 528

1491 TCGA-KIRC, 199 of 200 CPTAC-3, 263 of 269 TCGA-KIRP and 65 of 65 TCGA-KICH

1492 patients. Grade is recorded for 520 of 528 TCGA-KIRC and 200 of 200 CPTAC-3 patients.

1493 The M1 column gives the percentage of all patients who are M1. Patients recorded MX are

1494 treated as M0, and patients with AJCC stage IV disease and a T category below T4 are recoded

1495 as M1 (2 TCGA-KIRC, 12 CPTAC-3, 5 TCGA-KIRP and 0 TCGA-KICH patients). The figure

1496 in parentheses is the percentage M1 among patients with an explicitly assessed M category (494

1497 of 528, 70 of 200, 99 of 269 and 36 of 65).

1498 Nodal status was explicitly assessed in 259 of 528 TCGA-KIRC patients, 17 (6.6%) of them

1499 node-positive, and in 56 of 200 CPTAC-3 patients, 8 (14.3%) of them node-positive. It is not

1500 tabulated because the number of patients explicitly assessed is not recorded for TCGA-KIRP or

1501 TCGA-KICH.

Grade is missing for every TCGA-KIRP and TCGA-KICH patient at this data release and is not a standard measure for chromophobe tumours, so it did not enter the subtype covariate set. For TCGA-KIRP and TCGA-KICH, deaths are counted among the patients entering the adjusted models, which are the 263 of 269 TCGA-KIRP patients with a recorded T category and all 65 TCGA-KICH patients. Of all 269 TCGA-KIRP patients with module scores, 42 died. The non-feature read fraction is defined in the Methods and the Fig. 1c legend. IQR, interquartile range.

Table 2 | Module eigengenes associated with overall survival in TCGA-KIRC and their external replication in CPTAC-3.

(a) Discovery, TCGA-KIRC

Network	Module	Genes	n	Events	HR, principal specification (95% CI)	FDR, within network	FDR, all 29 modules	HR, clinical model (95% CI)	FDR, clinical model
lncRNA	blue	246	504	163	0.71 (0.60 to 0.83)	4.8e-04	4.4e-04	0.73 (0.62 to 0.85)	0.0015
lncRNA	greenyellow	108	504	163	1.35 (1.15 to 1.60)	0.0027	0.0033	1.31 (1.11 to 1.54)	0.0092
lncRNA	turquoise	438	504	163	1.24 (1.06 to 1.45)	0.039	0.043	1.26 (1.08 to 1.49)	0.024
protein-coding	green	670	502	165	0.62 (0.53 to 0.74)	2.2e-07	4.9e-07	0.69 (0.59 to 0.81)	3.8e-05
protein-coding	purple	216	502	165	1.28 (1.09 to 1.50)	0.016	0.018	1.24 (1.06 to 1.45)	0.036

(b) External replication, CPTAC-3 (199 patients, 32 events)

Network	Module	HR, 12-parameter model (95% CI)	p	FDR	HR, 6-parameter model (95% CI)	p	FDR	Direction as in discovery
lncRNA	blue	0.78 (0.55 to 1.11)	0.17	0.21	0.79 (0.57 to 1.10)	0.16	0.20	yes
lncRNA	greenyellow	1.59 (1.12 to 2.25)	0.0097	0.042	1.42 (1.07 to 1.90)	0.016	0.068	yes

Network	Module	HR, 12-parameter model (95% CI)	p	FDR	HR, 6-parameter model (95% CI)	p	FDR	Direction as in discovery
lncRNA	turquoise	1.11 (0.81 to 1.53)	0.51	0.51	1.10 (0.85 to 1.43)	0.46	0.46	yes
protein-coding	green	0.73 (0.51 to 1.04)	0.082	0.14	0.74 (0.54 to 1.03)	0.071	0.12	yes
protein-coding	purple	1.52 (1.08 to 2.14)	0.017	0.042	1.41 (1.04 to 1.92)	0.027	0.068	yes

1513 Hazard ratios (HR) are per 1 standard deviation of the module eigengene, computed on the
 1514 expression matrix residualised on the alignment metrics.

1515 The principal specification is the module eigengene with age, sex, T category, nodal status, M1,
 1516 ordinal grade, the ESTIMATE stromal and immune scores, and the three STAR alignment
 1517 metrics (non-feature read fraction, multimapping fraction and log10 assigned reads). The clinical
 1518 model is age, sex, T category, nodal status, M1 and ordinal grade.

1519 False discovery rates (FDR) are Benjamini-Hochberg, controlled within network across the 13
 1520 protein-coding or 16 lncRNA modules and, in the column so labelled, jointly across all 29
 1521 modules. The five modules listed meet a 5% within-network false discovery rate under the
 1522 principal specification, and the same five meet the joint threshold.

1523 The two discovery sample sizes differ because the networks exclude different connectivity
 1524 outliers and models are fitted on complete cases.

1525 In CPTAC-3, module membership and eigengene loadings were fixed in discovery, only the
 1526 module coefficient was re-estimated, and eigengenes were standardised within CPTAC-3. Panel
 1527 (b) therefore reports replication of an association, not the performance of a fixed prediction rule
 1528 (Table 3). The 12-parameter model uses the principal covariate set and carries 2.7 events per
 1529 parameter, below the threshold of 5 applied elsewhere. The 6-parameter model (module

1530 eigengene, age, sex, T category, M1 and ordinal grade) carries 5.3 and meets that threshold. Its
 1531 false discovery rates are controlled across the five modules tested.
 1532 Standardising CPTAC-3 with the discovery means and standard deviations in place of within-
 1533 cohort standardisation changes no hazard ratio in panel (b) by more than 0.01 (Supplementary
 1534 Table 21g).

1535 **Table 3 | Performance of the locked prediction model for overall survival.**

1536 *(a) Discrimination*

Cohort	Comparator	Model	Standardisation	n	Events	C (95% CI)
TCGA-KIRC (apparent)	clinical	comparator only cohort		493	159	0.766 (0.729 to 0.804)
TCGA-KIRC (apparent)	clinical	+ module eigengenes cohort		493	159	0.787 (0.752 to 0.822)
TCGA-KIRC (cross-validated)	clinical	comparator only cohort		493	159	0.758 (0.720 to 0.796)
TCGA-KIRC (cross-validated)	clinical	+ module eigengenes cohort		493	159	0.767 (0.730 to 0.804)
TCGA-KIRC (cross-validated)	clinical	+ hub lncRNAs selected inside each fold cohort		493	159	0.777 (0.742 to 0.813)
CPTAC-3	clinical	comparator only cohort		199	32	0.750 (0.657 to 0.843)
CPTAC-3	clinical	+ module eigengenes cohort		199	32	0.759 (0.671 to 0.847)
CPTAC-3	clinical	comparator only discovery-fixed		199	32	0.750 (0.657 to 0.843)
CPTAC-3	clinical	+ module eigengenes discovery-fixed		199	32	0.754 (0.663 to 0.844)
TCGA-KIRC (apparent)	augmented	comparator only cohort		493	159	0.775 (0.738 to 0.812)

Cohort	Comparator	Model	Standardisation	n	Events	C (95% CI)
TCGA-KIRC (apparent)	augmented	+ module eigengenes	cohort	493	159	0.792 (0.757 to 0.826)
CPTAC-3	augmented	comparator only	cohort	199	32	0.730 (0.642 to 0.818)
CPTAC-3	augmented	+ module eigengenes	cohort	199	32	0.745 (0.650 to 0.839)

1537 (b) Incremental value over the comparator

Estimate	Comparator	Change in C-index	95% CI	p	Spread across repeats or assignments	n	Events
Apparent, discovery, locked model scored in sample	clinical	+0.0205	0.0022 to 0.0400	0.026	not applicable	493	159
Apparent, discovery, locked model scored in sample	augmented	+0.0165	-0.0005 to 0.0347	0.056	not applicable	493	159
Ten repeats of ten-fold cross-validation, module eigengenes, modules fixed on the full cohort	clinical	+0.0092	-0.0072 to 0.0261	0.27	mean over repeats +0.0056; range -0.0051 to 0.0151; 9 of 10 repeats positive	493	159
Ten repeats of ten-fold cross-validation, module eigengenes, modules fixed on the full cohort	augmented	+0.0105	-0.0071 to 0.0288	0.22	mean over repeats +0.0102; range 0.0046 to 0.0198; 10 of 10 positive	493	159
Ten repeats of ten-fold cross-validation, hub lncRNAs selected inside each fold	clinical	+0.0197	0.0020 to 0.0368	0.033	mean over repeats +0.0073; range -0.0080 to 0.0251; 8 of 10 positive	493	159
Ten repeats of ten-fold cross-validation, hub lncRNAs selected inside each fold	augmented	+0.0033	-0.0117 to 0.0188	0.67	mean over repeats -0.0103; range -0.0295 to 0.0077; 2 of 10 positive	493	159
Cross-validation with both networks and all loadings rebuilt inside each training fold, 3 folds by 5 assignments	clinical	+0.0059	not estimated	not estimated	SD across assignments 0.0106; range -0.0089 to 0.0175	493	159

Estimate	Comparator	Change in C-index	95% CI	p	Spread across repeats or assignments	n	Events
Cross-validation with both networks and all loadings rebuilt inside each training fold, 3 folds by 5 assignments	augmented	+0.0123	not estimated	not estimated	SD across assignments 0.0145; range - 0.0081 to 0.0277	493	159
External validation, CPTAC-3, locked model, cohort standardisation	clinical	+0.0087	-0.0331 to 0.0478	0.68	not applicable	199	32
External validation, CPTAC-3, locked model, discovery-fixed standardisation	clinical	+0.0039	-0.0383 to 0.0433	0.88	not applicable	199	32
External validation, CPTAC-3, locked model, cohort standardisation	augmented	+0.0147	-0.0192 to 0.0478	0.37	not applicable	199	32

1538 (c) Calibration in CPTAC-3, clinical comparator, eigengenes standardised within cohort

Model	Years	Calibration slope (95% CI)	Observed risk	Expected risk	Observed / expected	n at risk
clinical	3	0.97 (0.60 to 1.34)	0.144	0.192	0.75	95
clinical + modules	3	0.89 (0.58 to 1.20)	0.144	0.194	0.74	95
clinical	5	0.97 (0.60 to 1.34)	0.263	0.292	0.90	23
clinical + modules	5	0.89 (0.58 to 1.20)	0.263	0.295	0.89	23

1539 (d) Brier score, clinical comparator, eigengenes standardised within cohort

Cohort	Years	Clinical	Clinical + modules	Null
TCGA-KIRC	1	0.0715	0.0675	0.0850
TCGA-KIRC	3	0.1443	0.1386	0.1779
TCGA-KIRC	5	0.1935	0.1795	0.2319
CPTAC-3	1	0.0578	0.0607	0.0536
CPTAC-3	3	0.0995	0.0979	0.1260
CPTAC-3	5	0.1392	0.1455	0.1950

1540 The clinical comparator is age, sex, T category, nodal status, M1 and ordinal grade. The

1541 augmented comparator adds the ESTIMATE stromal and immune scores and the three STAR

1542 alignment metrics. Module eigengenes are the 29 discovery module eigengenes (13 protein-
 1543 coding and 16 lncRNA), entered into a penalised Cox model with the comparator terms
 1544 unpenalised.

1545 Rows marked apparent in panel (a), and the first two rows of panel (b), are scored on the patients
 1546 the locked model was fitted to. Rows marked cross-validated are the repeat-averaged out-of-fold
 1547 predictors of ten repeats of ten-fold cross-validation, so the optimism of the clinical comparator
 1548 is the difference between 0.766 and 0.758. CPTAC-3 rows apply the discovery coefficients
 1549 unchanged.

1550 Cohort standardisation centres and scales the module eigengenes within the cohort being scored.
 1551 Discovery-fixed standardisation applies the discovery constants, a fixed rule that does not use the
 1552 validation population.

1553 In panel (b), the change in C-index for a cross-validated row is the increment of the out-of-fold
 1554 predictor averaged over repeats, the quantity to which the confidence interval and p value
 1555 belong. The mean over individual repeats is a different quantity, given with its range in the
 1556 spread column. That range is not a confidence interval. Confidence intervals and p values are
 1557 from a paired patient bootstrap over 2,000 resamples. The fold-wise rows report the mean,
 1558 standard deviation and range over five random fold assignments and carry no bootstrap interval.

1559 In panel (c), absolute risks use the baseline hazard fixed in discovery. The calibration slope is the
 1560 coefficient of the linear predictor in a Cox model and does not depend on the horizon, so the
 1561 same slope appears at three and five years. The n at risk is the number of patients still at risk at
 1562 that horizon, not the analysis sample size. Under discovery-fixed standardisation the combined
 1563 model over-predicts risk roughly twofold, with an observed over expected ratio of 0.55 at three
 1564 years (Supplementary Table 21b, Supplementary Fig. 12b).

1565 In panel (d), lower is better, the null model assigns every patient the Kaplan-Meier risk, and
1566 Brier scores are weighted by the inverse probability of censoring. TCGA-KIRC rows are
1567 apparent (in-sample) scores. No interval is computed for a Brier score, so differences between
1568 models should not be read as a ranking. The largest difference in CPTAC-3 is 0.0063, at five
1569 years. In CPTAC-3 the combined model is worse than the clinical model at one and five years,
1570 and at one year both models are worse than the null.